

Course 27026 – Molecular biology – E2021

Exam number:

27026 Molecular Biology

Written exam, December 8th 2021,

Duration: 4 hours

All aid is allowed except personal communication and internet access.

Answers must be written in the boxes in the word document, and not exceed the provided space. Remember to write exam number at the top. Upload your answers as a pdf file.

The exam is divided into three parts, A, B and C. Part A is 20 Multiple Choice questions; part B is 8 short questions, while part C is one large problem with 10 associated questions. A, B and C counts for 30 %, 32 % and 38 % of the final grade.

The Multiple-Choice part is accessed separately in the Multiple-Choice Module at DE. Each question counts 1.5 % of the total grade.

Part A. Multiple-Choice (30 % - 1.5 % for each question)

The Multiple-Choice part is accessed in the Multiple-Choice Module at DE.

Part B. Short questions (32 %).

Problem B1. (4 points).

Why does a high-fidelity DNA polymerase contain a 3'-OH to 5'-OH exonuclease activity? How do cells benefit from this activity?

A high-fidelity DNA polymerase contains a **3'→5' exonuclease activity** because this serves as a **proofreading mechanism** during DNA replication. When the polymerase inserts an incorrect nucleotide, the mismatch causes the polymerase to stall, allowing the newly synthesized strand to shift into the exonuclease site, where the incorrect base is removed from the **3' end**. After excision, the polymerase resumes accurate 5'→3' synthesis.

Cells benefit greatly from this activity because it **reduces replication errors by several thousand-fold**, preventing mutations and maintaining genome stability. Without this **proofreading function**, cells would accumulate harmful mutations much more rapidly.

Problem B2. (4 points).

Why is it necessary to decatenate e.g. the loops of eukaryotic chromosomes, and what enzyme is responsible for decatenation of eukaryotic chromosomes?

Decatenation means **separating two interlinked DNA molecules**. When chromosomes are replicated, the two sister chromatids often become physically **catenated (linked like rings in a chain)** because the DNA helices twist around each other during replication. These DNA links must be removed before the chromatids can segregate properly during mitosis.

During eukaryotic DNA replication, the unwinding of the double helix and the progression of replication forks naturally introduce **topological stress**, causing the newly replicated sister chromatids to become entangled. If these catenanes are not resolved, the chromatids cannot be pulled apart during anaphase. This leads to chromosome breakage, mis-segregation, aneuploidy, and genomic instability.

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The enzyme responsible for resolving (decatenating) these interlinks in eukaryotes is **Topoisomerase II (Topo II)**. Topo II can cut **both strands** of one DNA duplex, pass another duplex through the break, and then reseal it. This action effectively untangles and separates the intertwined sister chromatids, ensuring they can be correctly partitioned into daughter cells. In summary, **decatenation is essential for proper chromosome segregation**, prevents DNA damage, and is carried out in eukaryotes by **Topoisomerase II**, the enzyme specialized for removing DNA catenanes.

Problem B3. (4 points). Translation initiation in eukaryotes.

Describe the role of the three listed initiation factors in eukaryotic translation and state which complex they are part of.

eIF4E	Role: eIF4E is the cap-binding protein that recognizes and binds the 5' m⁷G cap of the mRNA. This binding is essential for recruiting the translation machinery to the mRNA and initiating cap-dependent scanning. Complex: eIF4E is part of the eIF4F complex , together with eIF4G (a scaffold protein) and eIF4A (an RNA helicase).
eIF1	Role: eIF1 promotes start codon selection accuracy . It binds to the 40S subunit and helps maintain the small ribosomal subunit in an “open” scanning conformation, ensuring that translation begins at the correct AUG in a good Kozak context. Complex: eIF1 is part of the 43S pre-initiation complex (PIC) , which also includes the 40S subunit, eIF2–GTP–Met–tRNA _i , eIF3, and eIF5.
eIF3	Role: eIF3 is a large, multi-subunit initiation factor that prevents premature joining of the 60S subunit , stabilizes the 43S pre-initiation complex, and promotes mRNA recruitment via interactions with eIF4G. It also helps the 40S subunit bind the 5' end of mRNA. Complex: eIF3 is a central component of the 43S pre-initiation complex and also interacts with the 48S scanning complex once mRNA is recruited.

Problem B4. (4 points). Eukaryotic RNA polymerases

RNA polymerase II carboxyterminal domain (CTD) is characterized by several YSPTSPS-repeats, that are phosphorylated and dephosphorylated during transcription. The amino acids in the repeats are numbered from 1 to 7. Answer the following four questions:

1. Which amino acid(s) is/are phosphorylated?
2. What is the phosphorylation state when the RNA polymerase is recruited by TFIID to the promoter?
3. One of the phosphorylations is coupled to splicing and polyadenylation. Which?
4. How is phosphorylation/dephosphorylation coupled to transcription termination?

1. Which amino acid(s) is/are phosphorylated?

In the **YSPTSPS** heptad repeat, the amino acids that undergo phosphorylation are:

- **Serine 2 (Ser2)**
- **Serine 5 (Ser5)**
- **Serine 7 (Ser7)**

Tyrosine 1 (Tyr1) and Threonine 4 (Thr4) can also be phosphorylated under certain conditions, but **Ser2, Ser5, and Ser7** are the major and functionally central phosphorylation sites during the transcription cycle.

2. What is the phosphorylation state when RNA polymerase II is recruited by TFIID to the promoter?

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When RNA polymerase II is first recruited to the promoter by TFIID (and the general transcription factors), the CTD is **hypophosphorylated**, meaning it is essentially **unphosphorylated**. This unphosphorylated state (the **Pol IIa** form) is required for stable interaction with the initiation machinery at the promoter.

3. One of the phosphorylations is coupled to splicing and polyadenylation. Which?

Serine 2 phosphorylation (Ser2-P) is the modification associated with **co-transcriptional RNA processing**, including:

- **splicing**
- **polyadenylation of the 3' end**

As elongation progresses, Ser2-P becomes dominant along the gene body and especially near the 3' region, coordinating mRNA maturation.

4. How is phosphorylation/dephosphorylation coupled to transcription termination?

As Pol II approaches the end of the gene, there is a shift from Ser5-P dominance to **high Ser2-P**, followed by **CTD dephosphorylation**, which is essential for termination.

CTD phosphatases (such as **FCP1** and **Ssu72**) remove phosphates from Ser2 and Ser5, allowing:

- release of Pol II from the DNA
- dissociation of the transcription machinery
- **recycling** of Pol II into the hypophosphorylated Pol IIa form, ready for another transcription cycle

Thus, termination is tightly coupled to **progressive dephosphorylation** of the CTD.

Problem B5. (4 points). Shuttle vectors.

What is a shuttle vector? Describe the genetic elements that must be present in a shuttle vector intended for use in both *Escherichia coli* and *Saccharomyces cerevisiae*.

A **shuttle vector** is a plasmid designed to function in **two different organisms**, allowing DNA to be cloned in one host (usually *E. coli*) and then transferred to another (such as *S. cerevisiae*) for expression or analysis. To work in both systems, the vector must carry the essential genetic elements required for replication and selection in each host.

In *E. coli*, the shuttle vector needs a **bacterial origin of replication** (e.g., ColE1 ori) and a **bacterial selectable marker**, typically an antibiotic resistance gene such as *ampR*. These allow the plasmid to replicate and be maintained during cloning.

For use in *S. cerevisiae*, the vector must include a **yeast origin of replication**, such as an **ARS** (and sometimes a **CEN** or **2 μ** element for stability or high copy number), as well as a **yeast selectable marker** like *URA3*, *LEU2*, *TRP1*, or *HIS3*. These ensure replication and plasmid maintenance inside yeast.

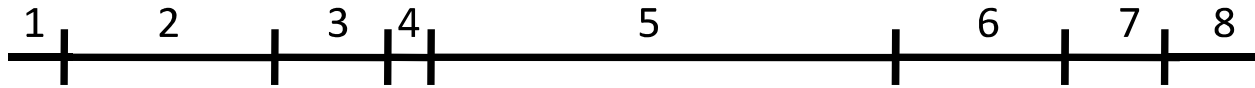
Most shuttle vectors also feature a **multiple cloning site** for inserting DNA and, if needed, a **yeast promoter** for gene expression.

In summary, a shuttle vector works in both *E. coli* and yeast because it carries **two origins of replication** and **two selectable markers**, one set for each organism.

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Problem B6. (4 points). mRNA processing

The drawing below shows the pre-mRNA from a eukaryotic gene. Element 1 encodes a signal sequence, whereas 2, 4, 6, and 8 are exons, and 3, 5, and 7 are introns. Describe the mature mRNA including the 3' and 5' end modifications. In which of the 8 elements will you find the start codon?



In the mature mRNA, all **introns (elements 3, 5, and 7)** have been removed by the spliceosome, so the transcript contains only the exons **1 → 2 → 4 → 6 → 8** joined seamlessly in that order. Immediately after transcription begins, the 5' end of the pre-mRNA is modified by addition of the **5' m⁷G cap**, which protects the RNA from degradation and is required for ribosome recruitment. At the opposite end, after transcription passes the polyadenylation signal, the RNA is cleaved and receives a **poly(A) tail**, which stabilizes the transcript and assists in export and translation.

Element **1** encodes a **signal sequence**, meaning it is part of the coding region that directs the nascent protein to the secretory pathway. However, the **first start codon (AUG)** that initiates translation is expected to lie within the **first coding exon**, which is **element 2**. This is because element 1 contains the *signal sequence coding region* itself, but the **start codon must appear at the beginning of this coding sequence**, and in typical eukaryotic gene organization, the AUG is found at the start of the first coding exon *after* the untranslated leader. Since element 1 is described as encoding the signal sequence, the AUG must initiate that region, and thus it resides in the **next exon**—element **2**.

So, the mature mRNA consists of:

5' cap – exon 1 – exon 2 – exon 4 – exon 6 – exon 8 – poly(A) tail,
and the **start codon is found in element 2**.

Problem B7. (4 points).

Describe and name two mechanisms used by host cells to defend themselves against viral attacks.

1. The Interferon-Mediated Antiviral State

One of the most powerful and immediate antiviral defenses is the **interferon response**, which is activated when a host cell detects viral molecules such as double-stranded RNA or unusual RNA structures produced during viral replication. These pathogen-associated patterns trigger intracellular sensors like RIG-I and MDA5, leading the infected cell to secrete **type I interferons** (IFN- α and IFN- β). These interferons bind receptors on both the infected cell and neighboring cells and induce the expression of a large set of **interferon-stimulated genes** that together create an antiviral environment. Among the most important effectors are PKR, which inhibits translation by phosphorylating eIF2 α and thereby prevents viral protein synthesis, and the OAS/RNase L system, which degrades viral RNA in the cytoplasm. Through these and many additional gene products, interferons slow or stop viral replication, limit viral spread, and alert immune cells to the infection. This coordinated response provides a rapid, broad-spectrum antiviral defense even before adaptive immunity is activated.

2. RNA Interference (RNAi) and siRNA-Guided Viral RNA Destruction

Another major antiviral mechanism is **RNA interference**, which allows cells to selectively destroy viral RNA molecules. When a virus generates double-stranded RNA or replication intermediates inside a host cell, the enzyme **Dicer** recognizes these as foreign and cleaves them into small interfering RNAs (siRNAs). These siRNAs are then loaded into the **RNA-induced silencing complex (RISC)**, where the Argonaute protein uses the siRNA as a guide to seek out complementary viral RNA sequences. Once bound, Argonaute cuts the viral RNA, preventing it from being translated or replicated. This mechanism provides a highly specific and efficient way for the cell to eliminate viral genomes and transcripts. RNAi is

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especially important in plants and invertebrates, but similar pathways also function in mammals and contribute to antiviral defense.

3. Apoptosis as an Antiviral Defense

A powerful cellular defense against viral infection is **programmed cell death**, or apoptosis. When a cell senses that it has been infected—often through detection of viral nucleic acids or abnormal viral protein accumulation—it can activate pro-apoptotic pathways involving p53, caspases, and mitochondrial signaling. By triggering apoptosis, the cell sacrifices itself to prevent the virus from completing its replication cycle and spreading to neighboring cells. Many viruses rely on the host cell's metabolic and replication machinery to propagate, so prematurely killing the infected cell dramatically limits viral production. Apoptosis also alerts the immune system, as dying cells release signals that recruit phagocytes and activate antigen-presenting cells. Although viruses have evolved proteins to block apoptosis, the host's ability to eliminate infected cells remains one of the most essential antiviral barriers.

4. Restriction Factors (Intrinsic Immunity), such as APOBEC3G and TRIM5α

Host cells also express **restriction factors**, a class of antiviral proteins that act directly on viral components to block infection at specific steps. For example, **APOBEC3G** is a cytidine deaminase that becomes incorporated into retroviral particles and mutates viral DNA during reverse transcription, rendering the viral genome nonfunctional. Another example, **TRIM5α**, recognizes the capsid of certain retroviruses immediately after entry and triggers its premature disassembly, preventing productive infection. Restriction factors are part of the cell's **intrinsic immunity**—a constitutively present defense system that does not require prior exposure to a pathogen. Because these proteins act very early in the viral life cycle and are often highly specific, they form an effective natural barrier to infection, especially against retroviruses and other RNA viruses.

Problem B8. (4 points).

Describe one mechanism by which bacteria can acquire resistance against antibiotics?

One important mechanism by which bacteria can acquire antibiotic resistance is **horizontal gene transfer**, where a bacterium gains resistance genes from another organism rather than evolving them internally. A common form of this is **conjugation**, in which bacteria transfer plasmids—small, circular DNA molecules—through direct cell-to-cell contact. Many plasmids carry **antibiotic resistance genes**, such as those encoding enzymes that degrade antibiotics (e.g., β -lactamases), efflux pumps that expel the drug, or modified targets that prevent the antibiotic from binding. When a bacterium receives such a plasmid, it immediately becomes resistant, and because plasmids can spread rapidly through a population—even between different bacterial species—resistance can disseminate very quickly. This mechanism is a major reason why antibiotic resistance can emerge and spread so efficiently in clinical and environmental settings.

OR

Another mechanism by which bacteria can acquire antibiotic resistance is through **spontaneous mutation in chromosomal genes** that alters the antibiotic's target. Antibiotics often work by binding to essential bacterial structures—such as ribosomal proteins, RNA polymerase, or enzymes involved in cell wall synthesis. A point mutation in the gene encoding one of these targets can change its structure just enough that the antibiotic can no longer bind effectively, while the enzyme or protein still retains its normal cellular function. For example, mutations in the gene encoding DNA gyrase can make bacteria resistant to fluoroquinolones, and mutations in ribosomal RNA can confer resistance to macrolides or aminoglycosides. Because these mutations

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arise naturally during DNA replication, even at low frequency, exposure to antibiotics strongly selects for the mutant cells, allowing them to become the dominant strain in the population

Part C. Regulation of gene expression in prokaryotes (38 %).

The expression of proteins A and B from *genA* and *genB* in *E.coli* is induced by adding an inducer to the growth medium. The enzymatic activity of A and B was assayed in the absence or presence of this inducer, and shown in table 1.

Table 1. The enzymatic activities of A and B assayed in the absence or presence of inducer.

Inducer addition	Enzymatic level protein A (units/mg)	Enzymatic level protein B (units/mg)
-	10	15
+	200	300

Question 1 (2 %). *genA* and *genB* were experimentally proven to constitute an operon. What is the criterion for *genA* and *genB* being in the same operon?

The criterion for **genA** and **genB** being in the **same operon** is that they show **coordinated expression**: both genes are induced together and to a similar degree in response to the **same inducer**, meaning their transcription is controlled by a **single promoter**.

In the data, adding the inducer increases protein A levels from **10 → 200 units** and protein B levels from **15 → 300 units**. This parallel induction indicates that the two genes are transcribed as part of **one polycistronic mRNA**, which is the defining characteristic of an operon in prokaryotes.

So the experimental criterion is:

Both genes change their expression simultaneously and proportionally in response to the same regulatory signal.

Question 2 (2 %). How can you experimentally verify that two genes are part of the same operon?

Answer:

To experimentally verify that two genes are part of the same operon you can verify that two genes belong to the same operon by showing that they are transcribed together on **one polycistronic mRNA**. This can be done by performing **RT-PCR across the region between the two genes**—if a single RNA molecule contains both sequences, you will amplify a product spanning the junction. Alternatively, a **Northern blot** showing that both gene-specific probes detect an mRNA of the same size also demonstrates co-transcription.

The genetic map of the *genAB* operon is shown in figure 1.

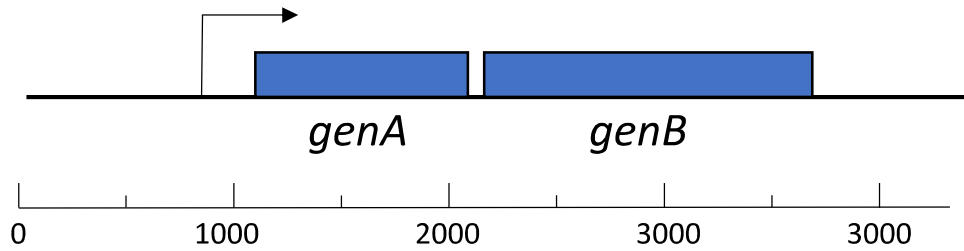


Figure 1. The genetic map of the *genAB* operon. An arrow indicates the approximate position of the promoter. A marker in basepairs below the map shows the size of the operon.

To identify the promoter, a set of transcriptional fusions between *genA* and *lacZ* were constructed, and introduced into *E. coli*. The cloned fragment ranges from 0 to 1200 base pairs according to the scale in Figure 1.

Question 3 (2 %). Before introducing the transcriptional fusions into the *E. coli* strain, the researcher inactivated one locus in the *E. coli* genome. Which *E. coli* gene needs to be inactivated in order to use the *lacZ* reporter fusions to measure promoter activity?

in short you can do this by inactivating either only the LacZ gene or the entire Lac operon.

To accurately measure promoter activity using a **lacZ reporter fusion**, the host *E. coli* strain must have its **chromosomal lacZ gene inactivated**. If native *lacZ* were still functional, the cell would produce its own β -galactosidase, creating **background enzyme activity** unrelated to the reporter construct. This background would interfere with measurements, because the experimenter would not be able to distinguish β -galactosidase produced from the **reporter fusion** from that produced by the cell's **endogenous lac operon**. By using a *lacZ*- strain (for example, a Δ *lacZ* or *lacZ*::Kan insertion mutant), any β -galactosidase detected can be attributed **exclusively** to the activity of the promoter driving the reporter, ensuring that the experiment reflects true regulatory changes rather than native expression noise.

What is reporter fusion?:

A **reporter fusion** is a tool that lets researchers monitor gene expression by linking a gene's regulatory region or coding sequence to an easily measurable **reporter gene**. The reporter produces a clear signal—such as color, fluorescence, or light—that reflects when and how strongly the original gene is being expressed.

The mechanism works by placing the reporter under the control of the gene's promoter (a **transcriptional fusion**) or fusing it directly to the gene's coding sequence (a **translational fusion**). When the promoter is active, or when the gene is translated, the reporter is produced along with it, allowing gene activity to be tracked indirectly through the reporter's signal.

Thus, reporter fusions function by using a **simple, detectable output** as a stand-in for the actual gene's expression, making it easy to study gene regulation and cellular responses.

After establishment of the transcriptional fusion, the β -galactosidase activity was assayed after growth in medium in the absence and presence of the inducer. The results are shown in Table 2.

Table 2. Enzymatic assay of *E. coli* strain carrying a *genA*::*lacZ* fusion

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Inducer addition	β -galactosidase activity (units/mg)
-	50
+	200

Question 4 (5 %). Comparing the induction of the enzymatic activities

Calculate i) the fold of induction in the enzymatic activities of A and B (Table 1), and ii) the fold of induction found in the transcriptional fusion (Table 2)	Answer: Induction fold: i) A/B enzyme assay = ii) <i>genA::lacZ</i> : =
If you find a difference, how would you explain it?	Answer:

To further characterize the regulation of the *genAB* operon, the *E. coli* strain harbouring the *genA::lacZ* fusion was subjected to transposon mutagenesis, and strains with altered *lacZ* expression were isolated. Two transposon mutants – Mut1 and Mut2 - were obtained. In Mut1 the transposon was inserted into a gene (*genP*) encoding a protein with several putative α -helices of hydrophobic nature. In Mut2, the transposon had inserted into *genR* encoding a protein with a characteristic Helix-Turn-Helix motif. Next, the β -galactosidase activity in Mut1 and Mut2 was determined in the absence and presence of the inducer. The measured activities are listed in Table 3.

Table 3. Enzymatic assay of *E. coli* strain carrying a *genA::lacZ* fusion

Strain	Mutant type	Inducer addition	β -galactosidase activity (units/mg)
Wildtype		-	50
		+	200
Mut1 (<i>genP</i>)	Hydrophobic α -helices	-	50
		+	50
Mut2 (<i>genR</i>)	Helix-Turn-Helix motif	-	200
		+	200

Question 5 (4 %).

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What does the presence of hydrophobic α -helices in a protein suggest about its function?	Answer:
How does the mutation in Mut1 alter transcription from the <i>genA</i> -promoter? Suggest a function for the protein mutated in Mut1. Briefly explain why.	Answer:

Question 6 (4 %).	
What does the presence of a Helix-Turn-Helix motif in a protein suggest about its function?	Answer:
How does the mutation in Mut2 alter transcription from the <i>genA</i> -promoter? Suggest a function for the protein mutated in Mut2. Briefly explain why.	Answer:

By PCR, *genR* was amplified from wildtype chromosomal DNA, and cloned into a plasmid to give pGenR⁺. This plasmid was subsequently transformed into the Mut2 strain. The resulting strain was assayed for β -galactosidase activity in the absence and presence of the inducer.

Question 7 (4 %). Predict the result of the β -galactosidase assay with the Mut1 carrying pGenR ⁺
Answer: Activity of Mut1//pGenR ⁺ without inducer: Activity of Mut1//pGenR ⁺ with inducer:

To further characterize the regulation of *genAB*, the promoter was substituted with a constitutive promoter, leaving the leader unchanged. The enzyme activity of A was measured (Table 4).

Table 4. The enzymatic activities of A and B assayed in a strain where the *genAB* operon is expressed from

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a constitutive promoter. Strain were grown either in the absence or presence of the inducer.

Inducer addition	Enzymatic level protein A (units/mg)
-	40
+	200

Question 8 (5 %). Compare the induction fold of the enzymatic activities

Calculate i) the fold of induction in the enzymatic activity of A (Table 1) and ii) the fold of induction found in the presence of a constitutive promoter (Table 4)

Answer:

Induction fold:

i) A enzyme assay wildtype promoter =

ii) A enzyme assay constitutive promoter =

If you find a difference, how would you explain it?

Answer:

The *genAB* leader can fold into two mutually exclusive structures shown in figure 2.

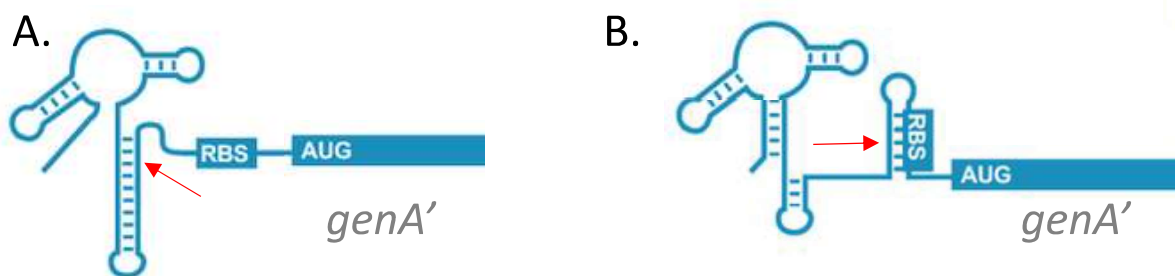


Figure 2. The untranslated leader in the *genAB* operon was predicted to be capable of folding into two alternative structures shown in A. and B. The Shine-Dalgarno sequence and the start codon are noted as RBS and AUG, respectively. The first part of the *genA* open reading frame is shown as a blue box. The red arrow points to the mutated nucleotide in Mut3. (See description of Mut3 in the text below).

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Question 9 (5 %). Comparing the structures of the leader (Figure 2)	
Structure A will result in higher translation initiation frequencies. Why?	Answer:
Which of the two structures will be stabilized by binding of the inducer? Explain why.	Answer:

A third mutant, Mut3, was obtained after UV mutagenesis of the strain expressing *genAB* from a constitutive promoter. The mutation in Mut3 resulted in constitutively high expression of *genAB*. The mutation in Mut3 introduced a base shift in the *genAB* leader at the position marked with a red arrow in figure 2.

Question 10 (5 %). Explain why Mut3 has a higher expression of <i>genAB</i> in the absence of the inducer.
Answer:

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Study number:

27026 Molecular Biology

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Duration: 4 hours

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Part A. Multiple-Choice (30 % - 1.5 % for each question)

The Multiple-Choice part is accessed in the Multiple-Choice Module at DE.

	correct answer
1. Which parameter significantly influences the stringency of primer binding in a PCR experiment amplifying a specific DNA sequence?	
A) Volume of the PCR reaction B) The concentration of deoxynucleotides (dNTPs) C) Annealing temperature D) DNA polymerase type E) Template DNA concentration	C Higher annealing temp needs more complementary -> higher stringency
2. What type of probe is typically used in a Southern blot?	
A) Sheared chromosomal DNA B) A fluorescence-labelled antibody C) A radioactive labeled oligo deoxyribonucleotide D) tRNA E) mRNA	C southern blot = DNA on the membrane + complementary DNA-probe (often radioactive)
3. Which biological process benefits from the presence of a telomere?	
A) Replication of centromeres B) Replication of chromosome ends C) Replication of plasmids D) Replication of exons E) Replication of introns	B telomere protects and enables replication of the ends (end-replication problem)
4. Replication of a chromosome requires RNA primers. In eukaryotes, what enzyme activity removes these primers to avoid RNA in the chromosomes?	
A) RNA exonuclease B) DNA Polymerase I exonuclease activity	E

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C) DNA Polymerase II exonuclease activity D) DNA Polymerase III exonuclease activity E) Flap endonuclease activity	FEN1 cleaves RNA-flaps after DNA pol δ displaced the primer
5. Why do you need topoisomerase activity in front of the replication fork?	
A) To remove replication-induced positive DNA supercoils B) To remove replication-induced negative DNA supercoils C) To remove the RNA primer D) To fuse Okazaki fragments E) To remove telomeres	
6. A yeast strain with defective telomerase may form circular chromosomes. Which DNA repair pathway is required to form the circles?	
A) Base-excision repair B) Nucleotide-excision repair C) Mismatch-excision repair D) Non-homologous End-joining E) Homologous recombination	
7. Formation of heteroduplex DNA may lead to ____?	
A) Gene conversion B) DNA crossing over C) DNA double-strand break formation D) DNA single-strand break formation E) Loss of telomeres	
8. Plasmids may transfer from one bacterial cell to another by conjugation, but what type of genetic material is transferred during conjugation?	
A) a linear single-stranded DNA molecule B) a linear double-stranded DNA molecule C) a circular single-stranded DNA molecule D) a circular double-stranded DNA molecule E) a linear RNA molecule	
9. Which additional step is needed to express a eukaryotic gene obtained by Reverse transcriptase PCR in a bacterium?	
A) Deletion of introns B) Removal of the 5'-Cap C) Removal of the polyA- tail D) Changing the U residues to T's E) Addition of Shine-Dalgarno sequence	
10. Which factor/event is NOT involved in eukaryotic transcription	
A) Phosphorylation of the Carboxy-terminal Domain (CTD) of the RNA polymerase B) TATA-box binding protein C) The initiator Inr D) The 43S preinitiation complex	

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E) Transcription factor TF _{II} D	
11. How is mRNA transcription termination typically achieved in eukaryotic cells?	
A) Polyadenylation signal recognition B) Release factor interaction C) Hairpin loop formation D) Rho protein termination E) tRNA dissociation	
12. Two snRNAs within the spliceosome forms the catalytic center responsible for the transesterification reactions during pre-mRNA splicing. Which?	
A) U1 and U4 snRNAs B) U2 and U5 snRNAs C) U2 and U6 snRNAs D) U4 and U6 snRNAs E) U5 and U6 snRNAs	
13. How is GTP hydrolysis associated with translation elongation?	
A) GTP hydrolysis is not required for translation elongation only initiation B) GTP hydrolysis is required for binding of the amino acid to the tRNA C) GTP hydrolysis is required for the formation of the peptide bond D) GTP hydrolysis is required for EF-Tu to dissociate from the ribosome E) GTP hydrolysis is required for binding of the release factor	
14. Which cellular structure is primarily responsible for the assembly of amino acids into a polypeptide chain during translation?	
A) Ribosome B) Mitochondria C) Endoplasmic reticulum D) Golgi apparatus E) Lysosome	
15. For most DNA-binding proteins the specificity of DNA binding is determined by direct contact with...	
A) Negatively charged phosphate backbone of DNA B) The major groove of DNA C) The minor groove of DNA D) The major and the minor groove of DNA at the same time E) No Direct contact with DNA, as binding requires specialized adaptors that interact with DNA	
16. In a eukaryote cell, a repressor...	
A) always binds to the promoter preventing the binding of RNA polymerase II B) is only interacting with true activators in the nucleus C) can control transcription by sequestering true activators in the cytoplasm D) cannot interact directly with DNA E) is only responding to environmental stimuli	

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17. In a prokaryote cell, a trans-regulatory element is...	
A) a non-coding DNA sequence B) always a protein that can be supplemented externally C) always encoded by the operon that it regulates D) a diffusible molecule that can be supplemented externally E) a TATA-less promoter	
18. What characterizes the terminator/antiterminator structures in bacterial genes regulated by attenuation?	
A) the terminator structure provokes the detachment of RNA polymerase and termination of mRNA synthesis B) the operon is controlled by repression C) the antiterminator structure is always stabilized by the binding of a metabolite D) the stability of the mRNA molecules depends on the recruitment of RNAses by the RISC complex E) the antiterminator structure provokes the detachment of RNA polymerase and termination of mRNA synthesis	
19. Which of the following statements is TRUE :	
A) Histone remodelling complexes are highly sequence-specific B) Histone remodelling complexes always condense the chromatin around the histones C) Histone acetylation is associated with repression of gene expression D) Histone remodelling complexes always contain an ATPase catalytic subunit E) Histone methylation is always associated with repression of gene expression	
20. Viruses are used as vectors for gene delivery to eukaryotic cells. Which of the following statements is FALSE	
A) Lentiviruses are suitable for stable gene expression B) Adenoviruses are suitable for transient gene expression C) Adenoviruses switch between lytic and lysogenic forms D) Lentiviruses integrate their genetic material within the host genome E) Adenoviruses have a broad cell tropism	

Part B. Short questions (34 %).

Problem B1. (4 %). Genomes

The end of the telomere is arranged as a T-loop as shown in the drawing below:

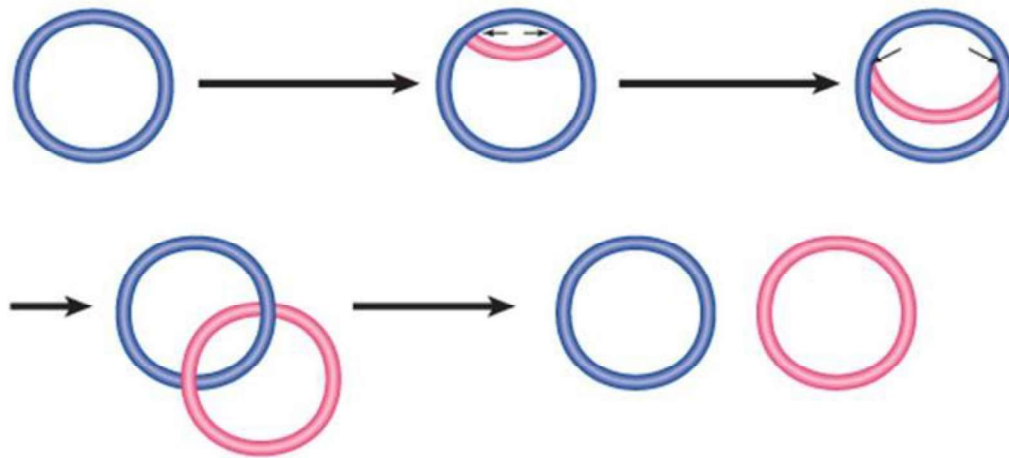


Explain why this is necessary?

Answer:

Problem B2. (6 %). Replication

The figure shows replication of the circular bacterial chromosome.



Which enzyme is required to separate the two chromosomes and what is the molecular mechanism?

Answer:

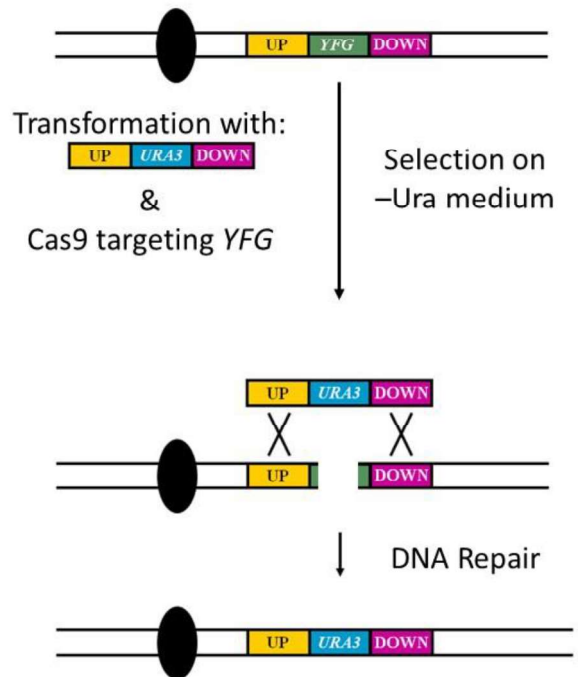
What would be the consequences if the two chromosomes remain catenated?

Answer:

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Problem B3. (6 %). Recombination

CRISPR technology has revolutionized strain engineering. The key feature of the technology is the Cas9 DNA endonuclease that can be guided to specifically cut any sequence in the genome. Many researchers who work with CRISPR technology in fungi prefer to use strains lacking the non-homologous end-joining DNA repair pathway. In the example to the right, Your Favorite Gene, *YFG* from yeast is deleted in a CRISPR experiment using a linear DNA fragment containing the selectable marker *URA3* flanked by homologous sequences up- and down-stream of *YFG*.



In the example above, explain which repair mechanism the yeast cell uses to repair the DNA double strand break induced by Cas9.

Answer:

Please elucidate the advantage of abolishing the non-homologous end-joining pathway when aiming to specifically delete a gene in a CRISPR experiment, considering the inevitability of random DNA double-strand breaks caused by cosmic irradiation and the existence of reactive oxidative molecules.

Answer:

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Problem B4. (6 %). The central dogma

A eukaryotic gene encodes a protein of 333 amino acids and analysis of the DNA sequence suggested the presence of two introns with a length of 500 and 200 bases, respectively. The size of the mature mRNA transcribed from the gene was experimentally estimated to be 1350 bases long. Assume that the size of the polyA tail is 250 bases and that the polyadenylation site is 50 bases after the stop codon.	
Calculate how many exons the gene encompasses?	Answer:
Calculate the approximate length of the leader	Answer:
Calculate the approximate length of the pre-mRNA	Answer:
Which method can be used to estimate the length of the mature mRNA (name only)	Answer:

Problem B5. (4 %). Transcription

In a study of the bacterial gene X that is transcribed from a constitutive promoter, three different mutants (M1, M2, and M3) showing reduced transcription of gene X were isolated. In the figure below, the DNA sequences of the gene X promoter region is shown for the wild-type and the three mutants (the first bases in the sequence were arbitrarily set to 1, and the identified mutations are shown in bold and by underscoring. The relative gene X mRNA levels in the strains are shown to the right, with the wild-type level being set to 100.

	1	10	20	30	40	50	Rel. mRNA
WT	C	C	G	T	A	T	100
M1	C	C	G	T	A	T	1
M2	C	C	G	T	A	T	1
M3	C	C	G	T	A	T	50

List a method that can be used to determine the relative mRNA levels (Name only)	Answer:
Write the first five bases of the gene X mRNA transcribed from the wildtype promoter	Answer:
Based on the sequences, how can you mechanistically explain the low transcription of gene X in mutant M1, M2, and M3?	Answer: M1:
	Answer: M2:
	Answer: M3:

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Problem B6. (4 %). Translation

Describe briefly the role of the three listed protein factors in eukaryotic translation initiation	
PABP	Answer:
eIF4E	Answer:
eIF3	Answer:

Problem B7. (4 %). Regulation

Explain the role of a boundary element (insulator) in eukaryotic gene regulation and chromatin remodeling.	Answer:
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Problem B8. (4 %). Virus

What is the difference between the multiplicity of infection (MOI) and the burst size of a certain bacteriophage?	Answer:
How is the multiplicity of infection (MOI) and the burst size of a bacteriophage calculated?	Answer:

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Part C. (36 %). The Histidine Biosynthesis Operon (*his* Operon) in *Escherichia coli*

The Histidine Biosynthesis Operon (*his* operon) in *Escherichia coli* controls the production of enzymes that synthesize the amino acid histidine. The expression of this operon is mainly regulated by the availability of cellular histidine.

The structure of the operon is as shown in the genetic map (fig 1).

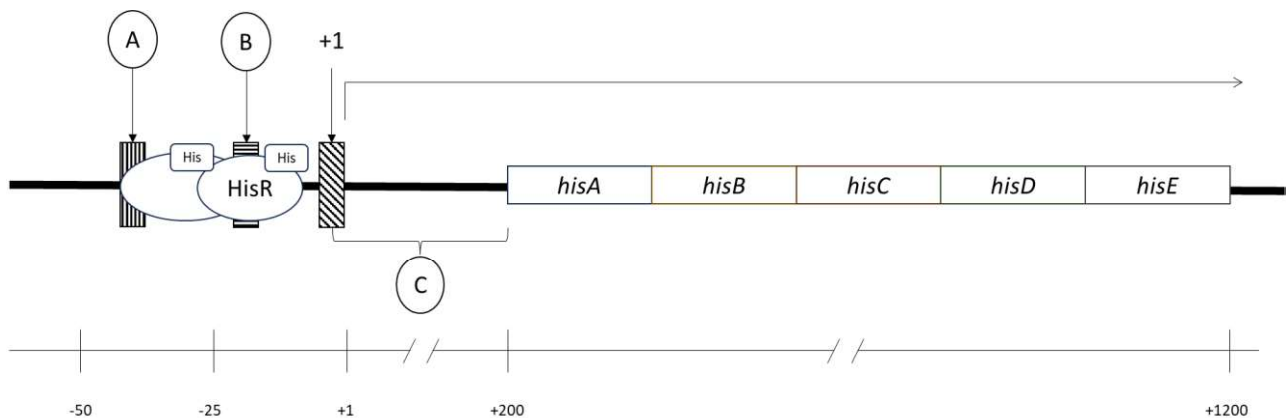


Figure 1. The genetic map of the *his* operon. The upper arrow symbolizes the mRNA produced from the operon, while the lower line serves as the reference ruler, designating +1 as the initiation point of transcription. The genes *hisA*, *hisB*, *hisC*, *hisD*, and *hisE* encode the enzymes—HisA, HisB, HisC, HisD, and HisE—critical for the synthesis of histidine. Expression of the *his* operon is controlled by HisR, that binds the amino acid Histidine and engages with the DNA in a dimeric structure. *A*, *B*, and *C* are addressed in the questions below.

Question 1 – (2%).

How would you classify the genes <i>hisA</i> , <i>hisB</i> , <i>hisC</i> , <i>hisD</i> , <i>hisE</i> , and <i>hisR</i> with respect to their function?	Answer:
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Question 2 – (4%).

Why is an operon structure possible in bacteria but not in eukaryotes?	Answer:
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Question 3 – (4%).

What specific type of DNA sequence would you expect that HisR-dimer to bind to?	Answer:
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Question 3 - (4%)

In Figure 1, name the elements A, B, and C.	Answer: A: B: C:
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A research team investigated the regulation of the *his* operon by growing wild-type bacteria and a set of mutants in the presence and absence of histidine and measuring expression of the *his* operon. To facilitate the readout, the genes *hisA*, *hisB*, *hisC*, *hisD*, and *hisE*, have been substituted by a *lacZ* reporter gene encoding the enzyme β -galactosidase.

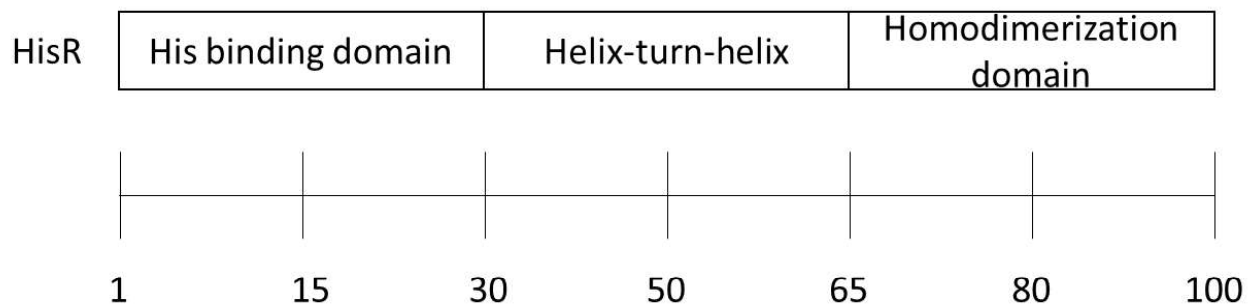


Figure 2. Domain structure of the protein HisR encoded by the gene *hisR*. The numbered scale indicates the number of the amino acids starting at the N-terminal end.

Table 1. Expression of the his operon measured as relative β -galactosidase activity (wild-type – histidine = 100 %). Activity of the β -galactosidase in the presence and the absence of Histidine. **Strain 1** is the wild-type strain, **Strain 2** has a complete deletion of the *hisR* gene, **Strain 3** has a complete deletion of *hisR*, and has been transformed with a plasmid encoding the HisR wild-type, **Strain 4** lacks residues 30 to 65 of HisR, **Strain 5** lacks the first 30 amino acids of HisR, **Strain 6** has a single point mutation in the *hisR* gene in which the Leucine (L) in position 80 has been substituted by a Serine (S).

		Relative β -galactosidase activity (%)	
Strain	Mutation	-Histidine	+Histidine
1	None (wild-type)	100	10
2	$\Delta hisR$	100	100
3	$\Delta hisR$ + plasmid- <i>hisR</i>	100	10
4	HisR- $\Delta 30-65$	100	100
5	HisR- $\Delta 1-30$	100	100
6	HisR-L80S	100	50

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Question 4 - (4%).

What kind of regulatory element (cis/trans) is encoded by <i>hisR</i> ? Briefly explain your response.	Answer:
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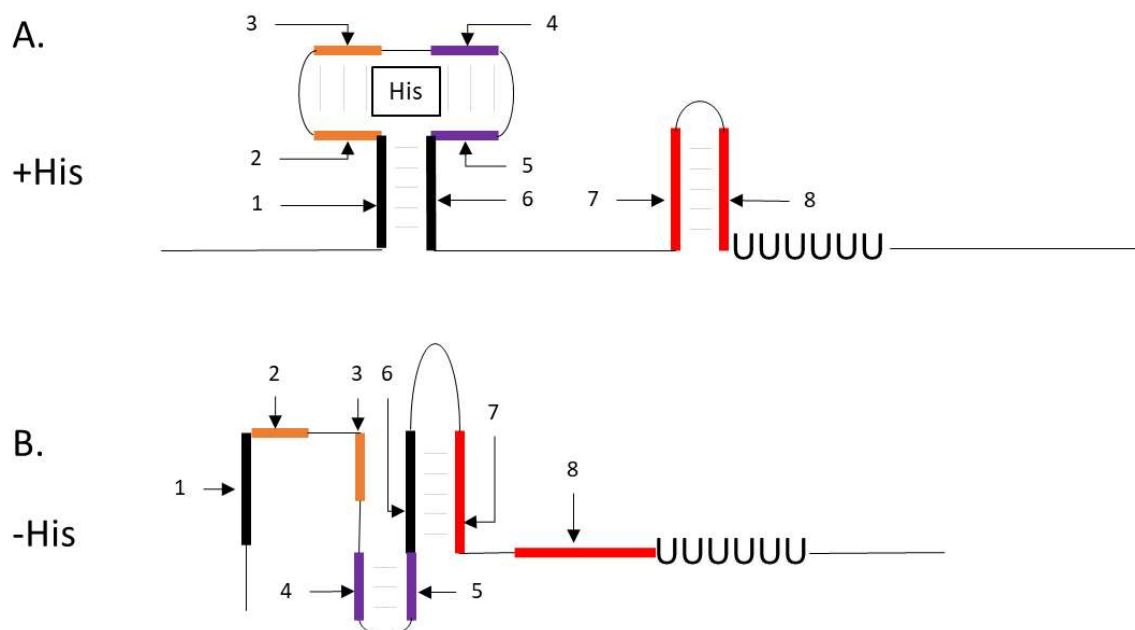
Question 5 – (6%).

How does HisR regulate the expression of the His operon in response to histidine availability in the wild-type? (hint: compare β -galactosidase activities measured in strain 1 and strain 2).	Answer:
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Question 6 – (6%).

Use Figure 2 to explain what functional domain of HisR that are deleted in strains 4, 5, and 6 and explain how the deletions in HisR proteins affect regulation of the <i>lacZ</i> reporter in the absence and presence of histidine in strain 4, 5 and 6. Use the relative activities of the three strains shown in Table 1.	Answer: Strain 4: Strain 5: Strain 6:
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The *his* operon has an extra layer of regulation: as depicted in Figure 3, the *his* mRNA leader sequence can adopt two mutually exclusive structures in the presence or absence of histidine.



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Figure 3. Mutually exclusive structures adopted by the mRNA leader sequence. This figure illustrates the alternative structures adopted by the mRNA leader sequence in two conditions: (A) in the presence of histidine (His) and (B) in the absence of histidine. Each fragment within the schematics is denoted by an identical number (1-8) in both representations (A and B). The grey parallel lines indicate the formation of hydrogen bonds between corresponding parallel fragments in the structures.

Question 7 - (2%).

What would you call the regulatory mechanism illustrated in figure 3?	Answer:
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Question 8 - (4%).

What are the names of the two different structures shown in Figure 3A and 3B, and how do folding of these two structures impact transcription of the <i>his</i> operon?	Answer:
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To better understand how the leader sequence contributes to regulation of the *his*-operon, the research team measured expression of β -galactosidase in strains (1-5) that all have the *lacZ* reporter gene fused to the same promoter, however, the strains differ by different deletions in the leader sequence. The strains were grown in the presence and absence of histidine and the results are shown below.

Question 9 – (4%).

Predict the activity of the reporter gene in the following table by filling in the missing numbers.			
Strain	Leader sequence	Relative β -galactosidase activity (%)	
		-Histidine	+Histidine
1	1-8 (wild-type)	100	10
2	$\Delta 8$		
3	$\Delta 6$		
4	$\Delta 1-6$		
5	$\Delta 7-8$		
Note 1: ΔX means that this particular DNA fragment has been deleted and is missing in the leader, that otherwise is intact. Also, the promoter and structural genes are as the wildtype.			
Note 2: $\Delta X-Y$ means that the DNA ranging between X and Y has been deleted.			

	correct answer
1. What kind of an enzyme is a restriction enzyme?	
A) Phosphatase B) Exonuclease C) Endonuclease D) Polymerase E) Phosphorylase Correct Answer: C) Endonuclease Restriction enzymes are endonucleases, meaning they cut DNA within the strand rather than removing nucleotides from the ends. They recognize specific short DNA sequences—typically 4–8 base pairs—and cleave the phosphodiester backbone at or near these sites. This sequence-specific internal cleavage is the defining property of restriction endonucleases. In bacteria, these enzymes form part of a restriction–modification system that protects the cell from foreign DNA: the bacterial genome is methylated at recognition sites, while unmethylated invading DNA (such as phage DNA) is cut and destroyed. In molecular biology, their predictable cutting patterns make them indispensable tools for cloning, mapping, and constructing recombinant DNA. A) Phosphatase — Incorrect Phosphatases catalyze the removal of phosphate groups from molecules, including nucleotides and proteins. Although useful in molecular biology—for example, alkaline phosphatase prevents vectors from self-ligating—they do not cut DNA strands, nor do they recognize DNA sequences. Because restriction enzymes function specifically by hydrolyzing phosphodiester bonds within DNA, a phosphatase cannot fulfill this role. B) Exonuclease — Incorrect Exonucleases degrade DNA by sequentially removing nucleotides from either the 5' or 3' end of a nucleic acid strand. This makes them fundamentally different from restriction enzymes, which cleave at internal positions rather than at the extremities. Since exonucleases require a free DNA end to act and restriction enzymes do not, this enzyme type does not match the activity of restriction endonucleases. D) Polymerase — Incorrect Polymerases synthesize nucleic acids by adding nucleotides to a growing DNA or RNA chain using a template strand. This anabolic function is the opposite of the degradative activity of restriction enzymes. Polymerases do not possess the ability to cleave DNA, recognize restriction sites, or break phosphodiester bonds, and therefore cannot be considered restriction enzymes. E) Phosphorylase — Incorrect	

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Phosphorylases catalyze phosphorolysis, a reaction in which inorganic phosphate is used to break chemical bonds, typically in carbohydrate metabolism. These enzymes do not hydrolyze DNA nor recognize specific nucleotide sequences. Because restriction enzymes use water, not inorganic phosphate, to cleave DNA and act with strict sequence specificity, phosphorylases cannot perform the function of a restriction enzyme.	
2. Which of the following statements regarding qPCR and RT-PCR is true?	
A) RT-PCR uses only one primer, qPCR uses two B) RT-PCR uses RNA as the primary template; qPCR uses DNA C) qPCR is not quantitative, RT-PCR is D) Only qPCR is used for generating DNA fragments for cloning E) RT-PCR uses NTP's as substrate – qPCR uses dNTP's Correct Answer: B) RT-PCR uses RNA as the primary template; qPCR uses DNA Reverse-transcription PCR (RT-PCR) begins with RNA as the starting material. In this method, the enzyme reverse transcriptase synthesizes complementary DNA (cDNA) from the RNA template. This cDNA then serves as the substrate for standard PCR amplification. In contrast, quantitative PCR (qPCR)—also called real-time PCR—amplifies DNA directly and measures the accumulation of PCR product in real time using fluorescent detection. Although qPCR can be combined with reverse transcription (RT-qPCR), qPCR as a technique fundamentally quantifies DNA amplification, not RNA directly. Thus, the statement correctly captures the essential distinction between RT-PCR (RNA → cDNA → amplification) and qPCR (DNA amplification measured quantitatively).	
A) RT-PCR uses only one primer, qPCR uses two — Incorrect Both RT-PCR and qPCR rely on two primers, a forward and reverse primer, during the PCR stage. Even in RT-PCR, after RNA is reverse-transcribed into cDNA, standard PCR conditions require a primer pair for amplification. Although a single primer is sometimes used during the reverse transcription step (e.g., oligo-dT or random hexamers), this is not the PCR stage itself. Therefore, stating that RT-PCR uses only one primer is incorrect.	
C) qPCR is not quantitative, RT-PCR is — Incorrect qPCR is specifically designed to be quantitative, measuring fluorescence during each amplification cycle to determine the initial amount of DNA present. It is the gold standard for quantitative gene expression analysis when used as RT-qPCR. RT-PCR, in its classical form, is not quantitative, because end-point PCR cannot reliably reflect starting template concentration. Thus, this option reverses the true properties of the techniques and is incorrect.	

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D) Only qPCR is used for generating DNA fragments for cloning — Incorrect qPCR can amplify DNA fragments, but it is not the standard method for producing DNA for cloning. Traditional PCR is commonly used to generate cloning fragments because it does not require fluorescent detection or real-time measurement. RT-PCR also produces DNA fragments (the cDNA product) that can be cloned. Therefore, the statement that only qPCR can be used for generating DNA cloning fragments is false.	
E) RT-PCR uses NTPs as substrate – qPCR uses dNTPs — Incorrect During the reverse transcription step of RT-PCR, the enzyme reverse transcriptase uses dNTPs, not NTPs, to synthesize cDNA. NTPs are used by RNA polymerases, not by reverse transcriptase. When PCR amplification follows (in either RT-PCR or qPCR), DNA polymerase always requires dNTPs, and qPCR is no exception. Thus, both RT-PCR and qPCR use dNTPs, making this statement incorrect.	
3. Which statement regarding RNAseq is true?	
A) RNAseq can only be used to quantify translated genes B) The number of reads is a direct measure of the gene expression C) Reverse transcription is a required part of the procedure D) qPCR is a required part of the procedure E) The procedure includes gel electrophoresis Correct Answer: C) Reverse transcription is a required part of the procedure RNA sequencing (RNAseq) begins with RNA as the primary biological material, but high-throughput sequencing platforms—such as Illumina—can sequence only DNA, not RNA. Therefore, a reverse transcription step is essential to convert RNA molecules into complementary DNA (cDNA). This cDNA is then used to construct sequencing libraries, which undergo amplification and ultimately sequencing. Without reverse transcription, the RNA would be incompatible with the sequencing chemistry, making this step an obligatory and fundamental component of every RNAseq workflow. Reverse transcription also stabilizes the genetic information, because cDNA is more chemically stable than RNA during library preparation.	
A) RNAseq can only be used to quantify translated genes — Incorrect RNAseq quantifies all RNA species present in the sample, including mRNA, but also non-translated RNA such as lincRNAs, antisense transcripts, precursor RNAs, and even small RNAs (with appropriate library preparation). It measures transcription, not translation. Translation is assessed using methods such as ribosome profiling, not standard RNAseq. Therefore, RNAseq is not limited to genes that are translated into proteins, making this statement incorrect.	
B) The number of reads is a direct measure of the gene expression — Incorrect (oversimplified) While read counts correlate with gene expression, they are not a direct measure because several factors influence read abundance: transcript length, sequencing depth, GC content biases, library preparation efficiency, and sample normalization. Therefore, raw read counts cannot be directly compared across genes or samples.	

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Proper quantification requires normalization methods such as FPKM, TPM, or DESeq2's size-factor normalization. Because the statement ignores these essential adjustments, it is considered incorrect.	
D) qPCR is a required part of the procedure — Incorrect qPCR is not required for RNAseq. It is often used as a complementary validation method to confirm differential expression results for selected genes, but it is not part of the RNAseq workflow itself. The sequencing data alone are sufficient for genome-wide expression profiling. Thus, qPCR plays an optional <i>validation</i> role, not a required procedural role, making this statement false.	
E) The procedure includes gel electrophoresis — Incorrect (not universally required) Gel electrophoresis is not an inherent or required step in RNAseq. Although gels or capillary electrophoresis instruments (e.g., Bioanalyzer) may be used to assess RNA integrity or library size distributions, these quality-control steps are not mandatory components of the RNAseq procedure itself. Modern workflows often use automated electropherograms rather than gels. Because electrophoresis is optional and not universally included, the statement is incorrect.	
4. The ends of the linear eukaryotic chromosomes are called...	
A) Monomers B) Dimers C) Multimers D) Polymers E) Telomeres Correct Answer: E) Telomeres The ends of linear eukaryotic chromosomes are called telomeres. Telomeres consist of repetitive DNA sequences—such as the canonical TTAGGG repeat in vertebrates—bound by specialized protein complexes that protect chromosomes from degradation, end-to-end fusion, and recognition as double-strand breaks. Because conventional DNA polymerases cannot fully replicate the extreme 5' ends of linear DNA (the end-replication problem), telomeres progressively shorten with each cell division in most somatic cells. The enzyme telomerase, a reverse transcriptase that carries an RNA template, counteracts this shortening in germ cells, stem cells, and certain cancer cells. Thus, telomeres are structurally and functionally distinct chromosome end regions, making option E the correct answer.	
A) Monomers — Incorrect A monomer is a single molecular unit that can join with others to form polymers. This term has no relevance to chromosome structure and does not describe the terminal regions of chromosomes. Chromosomes are complex assemblies of DNA and proteins, not monomeric units. Therefore, monomers cannot describe chromosome ends.	

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B) Dimers — Incorrect A dimer refers to a molecule or complex formed by two subunits, which may be identical (homodimer) or different (heterodimer). Dimers occur frequently in protein structure and transcription factor complexes, but they are unrelated to chromosome end regions. Telomeres are long tracts of repetitive DNA with associated proteins, not dimeric structures. Thus, this terminology is inappropriate for chromosome ends.	
C) Multimers — Incorrect A multimer describes a complex composed of multiple subunits, typically proteins or peptides. Although some telomeric proteins can oligomerize, the chromosome ends themselves are not referred to as multimers. The term is entirely outside the nomenclature of chromosomal architecture and does not describe DNA regions. Hence, this option is incorrect.	
D) Polymers — Incorrect While DNA itself is a polymer of nucleotides, the term “polymer” is far too general and does not designate a specific region of the chromosome. Every chromosome is made of polymeric DNA, but the specialized structures at the ends—telomeres—are defined by their repetitive sequences and protective proteins, not by their polymeric nature. Therefore, although technically true at a chemical level, “polymer” does not name the chromosome ends and is incorrect in this context.	
5. A pool of covalently double stranded DNA circles is composed by circles that are all negatively supercoiled. Among the possible topoisomers that each circle can form is...	
A) Double stranded circles with a bubble of single-stranded DNA B) Double stranded circles containing a nick C) Double stranded circles containing a double stranded break D) Double stranded circles containing a few uracil residues E) Double stranded circles containing a DNA replication fork Correct Answer: A) Double stranded circles with a bubble of single-stranded DNA A pool of covalently closed, negatively supercoiled DNA circles consists of molecules whose sugar–phosphate backbones are intact and whose negative superhelical tension is stored in the form of extra helical turns. One possible topoisomeric outcome of such molecules is the formation of localized denaturation bubbles—regions where the two DNA strands transiently separate without breaking the phosphodiester backbone. Negative supercoiling destabilizes the double helix and lowers the energetic cost of strand separation; thus, it promotes the formation of small single-stranded regions as an alternative way to absorb superhelical strain. These structures remain covalently closed, maintain topological integrity, and are	

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recognized as valid topological isomers (topoisomers) of negatively supercoiled DNA. Therefore, option A correctly describes a possible topoisomer of such circles.	
B) Double stranded circles containing a nick — Incorrect A nick—a single phosphodiester break in one DNA strand—allows free rotation around the intact strand. This instantly relaxes all supercoiling, because the linking number is no longer fixed. A nicked circle cannot maintain negative supercoils and therefore cannot exist as a topoisomer of a covalently closed supercoiled DNA pool. Although nicked DNA can appear experimentally, it is not a topoisomer of a covalently closed, supercoiled molecule.	
C) Double stranded circles containing a double-stranded break — Incorrect A double-stranded break converts the circular DNA into a linear molecule. Linear DNA has no defined linking number and cannot store supercoils. Because topology is defined only in covalently closed circular molecules, a double-strand break eliminates the topological state altogether. Thus, such a molecule cannot be considered a topoisomer of a supercoiled circle.	
D) Double stranded circles containing a few uracil residues — Incorrect The presence of uracil bases in DNA affects base composition, not DNA topology. Supercoiling depends on the integrity of the phosphodiester backbone and the imposed linking number, not on the identity of the bases. A circle with uracil residues would still have the same linking number and would not represent a distinct topoisomer. Therefore, this modification has no relevance to the topological possibilities of covalently closed supercoiled DNA.	
E) Double stranded circles containing a DNA replication fork — Incorrect A replication fork generates regions of ssDNA and can introduce topological strain, but it occurs only when DNA replication machinery is active. More importantly, a replication fork alters the covalent continuity of the DNA strands and introduces branched structures that are not considered simple topoisomers of a covalently closed circular DNA molecule. Topoisomers are defined for intact circles with fixed linking number, while a replication fork represents a specific biological intermediate, not a topological state available to every molecule in the supercoiled pool.	
6. Why are genomically integrated DNA transposons flanked by a short direct repeat of DNA?	
A) The transpose recognizes a short direct repeat of DNA in the genome; and this is where it inserts the transposon. B) The ends of the transposons are replicated twice just prior to be integrated into the genome	

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C) The transposase cleaves the host DNA to produce staggered DNA ends; and the single stranded ends of this break is the source of the direct repeat

D) During transposition, a DNA topoisomerase is recruited that is designed to create the direct repeat

E) The transposon contains a direct repeat in the ends of its own DNA, which it uses to ensure perfect replication of itself

Correct Answer: C) The transposase cleaves the host DNA to produce staggered DNA ends; and the single stranded ends of this break is the source of the direct repeat

Genomically integrated **DNA transposons** are characteristically flanked by short **target site duplications** (TSDs), which are short direct repeats of host DNA. These repeats do **not** originate from the transposon itself but from the **mechanism of transposase-mediated insertion**. During integration, the transposase enzyme cleaves the target DNA in a **staggered** fashion, producing short single-stranded overhangs on the host genome. When the transposon is inserted between these staggered cuts, DNA polymerase fills in the resulting single-stranded gaps, and DNA ligase seals them. Because the two staggered cuts were offset by a specific number of nucleotides, the filling-in process generates **duplicated sequences** on either side of the newly inserted transposon. These duplicated sequences form the direct repeats observed in integrated transposons. Therefore, the staggered cutting mechanism is the fundamental reason TSDs appear, making option C the correct answer.

A) The transposase recognizes a short direct repeat of DNA in the genome; and this is where it inserts the transposon — Incorrect

The transposase does not search for or recognize specific direct repeats. Many DNA transposons have minimal or relaxed target sequence specificity, and the repeated sequence arises **after** insertion, not before. The direct repeats are therefore not a preexisting signal for insertion but a byproduct of staggered cleavage and repair. Thus, the explanation in option A reverses causation and is incorrect.

B) The ends of the transposons are replicated twice just prior to being integrated into the genome — Incorrect

Transposon ends are not duplicated during their movement. The transposase cuts the transposon out of its original location (in cut-and-paste transposons) or replicates through it (in replicative transposons), but duplication of the **host** target sequence—not the transposon ends—is what produces the flanking repeats. Since no special double replication of transposon ends occurs before integration, option B is incorrect.

D) During transposition, a DNA topoisomerase is recruited that is designed to create the direct repeat — Incorrect

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Topoisomerases resolve supercoiling and DNA tangles; they do not create specific sequence duplications. They may participate in maintaining DNA topology during transposition, but they do not generate staggered cuts at defined offsets, nor do they synthesize new DNA needed to form direct repeats. The direct repeats arise through polymerase-mediated gap filling after staggered transposase cleavage, not through topoisomerase activity. Thus, this option is incorrect.	
E) The transposon contains a direct repeat in the ends of its own DNA, which it uses to ensure perfect replication of itself — Incorrect The ends of DNA transposons contain terminal inverted repeats (TIRs), not direct repeats. TIRs are recognized by the transposase and are essential for the transposition reaction. The direct repeats observed in the genome are not part of the transposon sequence but instead arise from duplication of host DNA during the insertion process. Therefore, the mechanism described in option E is incorrect both mechanistically and structurally.	
7. Lagging Synthesis is the consequence of ...	
A) Eukaryotic genomes being linear B) Prokaryotic genomes being circles C) DNA polymerases possessing only a 3'-OH to 5'-OH DNA synthesis activity D) DNA polymerases possessing only a 5'-OH to 3'-OH DNA synthesis activity E) DNA polymerases possessing both 3'-OH to 5'-OH and 5'-OH to 3'-OH DNA synthesis activities Correct Answer: D) DNA polymerases possessing only a 5'-to-3' DNA synthesis activity Lagging strand synthesis arises because all known DNA polymerases synthesize DNA exclusively in the 5'→3' direction. During DNA replication, the two strands of the double helix run antiparallel, meaning that while the leading strand can be synthesized continuously in the direction of the replication fork, the opposite strand is oriented 3'→5' relative to fork movement. Because polymerases cannot add nucleotides in the 3'→5' direction, the cell must synthesize the lagging strand discontinuously as short fragments—Okazaki fragments—each initiated by an RNA primer and later ligated together. Thus, the inherent directional constraint of DNA polymerases is the molecular reason why lagging strand synthesis occurs, making option D the correct answer.	
A) Eukaryotic genomes being linear — Incorrect The linearity of eukaryotic chromosomes does not cause lagging strand synthesis. Both linear and circular genomes require lagging synthesis because the issue arises from the directionality of polymerase, not from chromosome shape. Circular prokaryotic genomes also exhibit leading and lagging strands. Therefore, eukaryotic chromosome structure is irrelevant to the cause of lagging strand synthesis.	

B) Prokaryotic genomes being circles — Incorrect Circularity also does not dictate the existence of lagging synthesis. Even in circular genomes, replication forks generate antiparallel templates that require one strand to be synthesized continuously and the other discontinuously. The phenomenon is universal across organisms and independent of whether the genome is circular or linear. Thus, the structure of prokaryotic genomes does not explain lagging synthesis.	
C) DNA polymerases possessing only a 3'-OH to 5'-OH DNA synthesis activity — Incorrect This option describes a DNA synthesis polarity that does not exist in any known natural DNA polymerase. Polymerases cannot synthesize DNA in the 3'→5' direction because nucleotide addition requires a nucleophilic attack by the 3'-OH of the growing strand on the α-phosphate of an incoming dNTP. A hypothetical 3'→5' polymerase would not have an energetically favorable mechanism to maintain proofreading and phosphodiester bond formation. Therefore, this description is chemically incorrect and does not explain lagging strand synthesis.	
E) DNA polymerases possessing both 3'-OH to 5'-OH and 5'-OH to 3'-OH DNA synthesis activities — Incorrect No DNA polymerase possesses bidirectional synthesis capability. If polymerases could synthesize in both directions, there would be no need for lagging strand synthesis, as both template orientations could be copied continuously. Since lagging synthesis exists universally in living organisms, this option contradicts fundamental biochemical constraints of DNA polymerases and is therefore false.	
8. An exonuclease activity is essential during homologous recombination based repair of DNA double stranded breaks as it...	
A) Eliminates methylated nucleotides at the DNA DSB B) Eliminates pyrimidine dimers at the break C) Eliminates uracil residues at the break D) Creates 3'-OH single stranded ends that serves to identify identical template DNA E) Creates 5'-OH single stranded ends that serves to identify identical template DNA Correct Answer: D) Creates 3'-OH single stranded ends that serve to identify identical template DNA During homologous recombination–based repair of a DNA double-strand break (DSB), an exonuclease activity is essential for generating 3' single-stranded DNA (ssDNA) overhangs. When a DSB occurs, the broken DNA ends are first processed by nucleases that resect the 5' strands, thereby exposing 3'-OH ssDNA tails. These 3' overhangs are then coated by recombination proteins such as Rad51 (in eukaryotes) or RecA (in bacteria), which search the genome for a region of sequence homology. The 3' ssDNA tail subsequently invades the homologous duplex DNA,	

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initiating strand exchange and copying from the intact sister chromatid or homologous chromosome. Because homology recognition and strand invasion absolutely require a 3'-OH ssDNA overhang, the exonuclease activity responsible for producing it is indispensable. This makes option D the correct answer.	
A) Eliminates methylated nucleotides at the DNA DSB — Incorrect Removal of methylated nucleotides is not a function of homologous recombination enzymes. Methylation status does not interfere with the homology search or strand invasion steps, and no specific demethylation step is required for HR-mediated repair. Therefore, this explanation has no mechanistic relevance to the exonuclease activity involved in homologous recombination.	
B) Eliminates pyrimidine dimers at the break — Incorrect Pyrimidine dimers (such as thymine dimers caused by UV light) are repaired primarily by nucleotide excision repair (NER), not homologous recombination. Although severe damage can generate DSBs indirectly, the exonuclease in HR is not responsible for removing pyrimidine dimers. Instead, its sole role in HR is to produce resected 3' ends, so this option is incorrect.	
C) Eliminates uracil residues at the break — Incorrect Uracil in DNA is removed by base excision repair (BER) via uracil DNA glycosylase, not by the exonuclease involved in homologous recombination. Homologous recombination repair does not require the specific elimination of uracil; the exonuclease's function is structural (to generate ssDNA), not chemical (to remove inappropriate bases). Thus, this option is mechanistically unrelated and incorrect.	
E) Creates 5'-OH single stranded ends that serve to identify identical template DNA — Incorrect This option reverses the polarity. During HR, 5' ends are degraded, not generated. The essential recombination intermediate is a 3'-OH ssDNA overhang, because this 3' group acts as the nucleophile for DNA polymerase during strand extension once homology is found. A 5' ssDNA end cannot serve this function and would not support strand invasion or DNA synthesis. Therefore, creating 5'-OH single-stranded ends is not the purpose of the exonuclease, making this option incorrect.	
9. RNA polymerase III promoters – Which statement is FALSE?	
A) RNA polymerase III promoters express small RNAs B) In RNA polymerase III promoters TF_{III}C binding recruits TF_{III}B to the promoter. C) The TATAbox binding protein (TBP) is NOT part of TF_{III}B but only used in RNA polymerase II promoters D) RNA polymerase III promoter type 1 and type 2 has the BoxA as a common motif E) RNA polymerase III promoters are either Downstream or Upstream promoters	

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Correct Answer (the FALSE statement): C) The TATA-box binding protein (TBP) is NOT part of TFIIB but only used in RNA polymerase II promoters This statement is false because TBP is a core component of TFIIB, the general transcription factor required for RNA polymerase III (Pol III) transcription initiation. TFIIB consists of TBP, BRF1/BRF2, and BDP1, and TBP is essential for positioning Pol III at the transcription start site. While TBP is famous for its role in RNA polymerase II promoters (binding the TATA box), it also functions in RNA polymerase I and III transcription. Therefore, the claim that TBP is <i>not</i> part of TFIIB is incorrect, making option C the false statement.	
A) RNA polymerase III promoters express small RNAs — TRUE RNA polymerase III transcribes short, noncoding RNAs, including tRNAs, 5S rRNA, U6 snRNA, and various small structural RNAs. These RNAs are typically less than 300 nucleotides in length, so the statement accurately reflects Pol III's function. Thus, this option is true and not the false statement.	
B) In RNA polymerase III promoters TFIIC binding recruits TFIIB to the promoter — TRUE For type 1 and type 2 Pol III promoters (such as 5S rRNA and tRNA genes), TFIIC binds to internal promoter elements such as BoxA and BoxB, and this binding event recruits TFIIB upstream of the transcription start site. TFIIB then positions RNA polymerase III. Because this mechanism is central to Pol III transcription initiation, the statement is correct.	
D) RNA polymerase III promoter type 1 and type 2 has the BoxA as a common motif — TRUE Type 1 promoters (5S rRNA genes) contain BoxA, Intermediate Element (IE), and BoxC. Type 2 promoters (tRNA genes) contain BoxA and BoxB. Thus, BoxA is the shared conserved motif between type 1 and type 2 Pol III promoters. This statement is therefore true.	
E) RNA polymerase III promoters are either Downstream or Upstream promoters — TRUE Pol III promoters occur in two organizational classes: • Internal (downstream) promoters, where the promoter elements lie <i>within</i> the transcribed region—e.g., type 1 (5S rRNA) and type 2 (tRNA) promoters. • Upstream promoters, such as type 3 promoters (e.g., U6 snRNA), which resemble Pol II promoters and contain classical upstream elements like a TATA box. Because Pol III promoters are indeed categorized as downstream or upstream promoters, this statement is true.	
10. Some RNA polymerase II promoters do NOT include a TATA box, but instead they have...	
A) an Initiator B) a Pribnow box C) a – 35 sequence D) a BoxA element	

E) a downstream promoter element (DPE) Correct Answer: A) an Initiator Many RNA polymerase II (Pol II) promoters lack a TATA box and instead rely on other core promoter elements, the most common of which is the Initiator (Inr). The Initiator surrounds the transcription start site (TSS) and provides an alternative binding platform for the general transcription machinery, particularly TFIID. The Inr can function alone as the sole core promoter element in TATA-less promoters or in combination with other elements such as DPE (Downstream Promoter Element). Because the Initiator is the best-characterized and most widely used alternative to the TATA box in Pol II promoters, option A is the correct answer.	
B) a Pribnow box — Incorrect The Pribnow box (-10 element, TATAAT) is a bacterial promoter element recognized by σ-factor in prokaryotes. It is not found in eukaryotic RNA polymerase II promoters. The presence of a Pribnow box is strictly prokaryotic and has no role in eukaryotic transcription initiation. Therefore, this option is incorrect.	
C) a -35 sequence — Incorrect The -35 element is another bacterial promoter motif (TTGACA) recognized by σ-factor. Similar to the Pribnow box, it is part of the canonical prokaryotic promoter architecture and is absent from eukaryotic Pol II promoters. Thus, a -35 sequence cannot replace a TATA box in Pol II promoters, making this option incorrect.	
D) a BoxA element — Incorrect BoxA is a motif found in RNA polymerase III promoters, especially those controlling tRNA and 5S rRNA genes. It is an internal promoter element, not an upstream Pol II promoter component. Because BoxA is entirely unrelated to Pol II transcription initiation, it cannot substitute for a TATA box in Pol II promoters. Therefore, this option is incorrect.	
E) a downstream promoter element (DPE) — True but <i>not the best</i> answer The DPE is indeed a legitimate Pol II core promoter element that can function in the absence of a TATA box. However, the question asks for <i>the</i> element that TATA-less promoters have <i>instead</i>, and the canonical, primary alternative is the Initiator (Inr). The DPE typically requires an Initiator to function properly and is not usually described as the sole replacement for a missing TATA box. Thus, although DPE is associated with TATA-less promoters, it is not the primary element that replaces the TATA box, which makes option A the intended correct answer.	
11. Translation. Consider the following DNA sequence (Coding strand):	

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5'-ATGGCTGATATGCGCGCT-3'. Which peptide will it potentially encode (N-terminal to the left)?	
A) MADMRA B) WLIC C) GYAR D) SAV E) ARMDAM Correct Answer: A) MADMRA The given DNA sequence is the coding strand: 5'-ATGGCTGATATGCGCGCT-3' For a coding strand, the corresponding mRNA sequence is identical except that T is replaced by U: 5'-AUG GCU GAU AUG CGC GCU-3' We then read this mRNA in triplets (codons) from the 5' end, starting at the first AUG (which is both a start codon and codes for methionine, M): • AUG → M (Methionine) • GCU → A (Alanine) • GAU → D (Aspartate) • AUG → M (Methionine) • CGC → R (Arginine) • GCU → A (Alanine) This gives the peptide (from N-terminus to C-terminus): M-A-D-M-R-A → MADMRA Thus, option A) MADMRA is the correct answer.	
B) WLIC — Incorrect The peptide WLIC would correspond to codons such as: • W (Trp) → UGG • L (Leu) → e.g. UUA, UUG, CUA, CUG, etc. • I (Ile) → AUU, AUC, AUA • C (Cys) → UGU, UGC None of these codons appear in the translated mRNA AUG GCU GAU AUG CGC GCU. There is no UGG, no isoleucine codon, and no cysteine codon. Therefore, B does not match the codon sequence derived from the DNA.	
C) GYAR — Incorrect To obtain GYAR, we would need codons for: • G (Gly) → GGU, GGC, GGA, GGG • Y (Tyr) → UAU, UAC • A (Ala) → GCU, GCC, GCA, GCG • R (Arg) → CGU, CGC, CGA, CGG, AGA, AGG	

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Our actual codons are AUG GCU GAU AUG CGC GCU, encoding M, A, D, M, R, A. There is no glycine codon (GGN) and no tyrosine codon (UAU/UAC) present. Thus, C cannot be the correct peptide.	
D) SAV — Incorrect The peptide SAV would require codons such as: • S (Ser) → UCU, UCC, UCA, UCG, AGU, AGC • A (Ala) → GCU, GCC, GCA, GCG • V (Val) → GUU, GUC, GUA, GUG While alanine (A) is present in our sequence (GCU), we do not see any serine codons or valine codons in the codon series AUG GCU GAU AUG CGC GCU. Hence, D does not match the translation of the given coding strand.	
E) ARMDAM — Incorrect ARMDAM would correspond to: • A → Ala (GCU, GCC, GCA, GCG) • R → Arg (CGU, CGC, CGA, CGG, AGA, AGG) • M → Met (AUG) • D → Asp (GAU, GAC) • A → Ala • M → Met Our actual peptide is MADMRA, not ARMDAM. Even though the letters A, R, M, D appear in both, the order is strictly determined by the codon sequence read from the 5' end, and the sequence we get from the mRNA AUG GCU GAU AUG CGC GCU is unambiguously M–A–D–M–R–A. Thus, E is incorrect.	
12. Translation in pro- and eukaryotes share many common features, but some is still different. Which of the following processes is fundamentally different?	
A) GTP is used to energize ribosome translocation during the translation elongation process B) AUG is a start codon C) Aminoacyl-tRNA brings the amino acid into the A site of the ribosome D) The stop codons are recognized by release factors with a structure mimicking a tRNA E) The small subunit of the ribosome is recruited to the 5' end of the mRNA as part of the translation initiation process Correct Answer: E) The small subunit of the ribosome is recruited to the 5' end of the mRNA as part of the translation initiation process This process is fundamentally different between prokaryotes and eukaryotes. In eukaryotes, translation initiation is strictly 5' cap-dependent: the small ribosomal subunit (40S), together with initiation factors such as eIF4E, eIF4G, and eIF4A, binds the 5'-m⁷G cap of the mRNA and then scans along the mRNA to locate the first suitable AUG start codon (the scanning mechanism).	

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In prokaryotes, however, the ribosome does not bind the 5' end of the mRNA. Instead, the small subunit (30S) recognizes internal sites through the Shine-Dalgarno sequence, a ribosome-binding site upstream of the start codon that base-pairs with the 16S rRNA. Translation often begins independently of the 5' end, and polycistronic mRNAs contain multiple Shine-Dalgarno sequences. Because initiation site recognition is mechanistically and conceptually distinct in the two systems, option E correctly identifies the fundamentally different process.	
A) GTP is used to energize ribosome translocation during elongation — TRUE in both systems Both prokaryotic and eukaryotic translation elongation use GTP hydrolysis to drive ribosome translocation. • In bacteria, EF-G uses GTP. • In eukaryotes, its counterpart eEF2 uses GTP. Because this mechanism is conserved, it is not fundamentally different.	
B) AUG is a start codon — TRUE in both systems Both prokaryotes and eukaryotes initiate translation at AUG, encoding methionine (formyl-methionine in bacteria, methionine in eukaryotes). While context signals differ (Kozak sequence vs. Shine-Dalgarno), the use of AUG itself is conserved. Thus, this is not a distinguishing feature.	
C) Aminoacyl-tRNA enters the A site of the ribosome — TRUE in both systems In both domains of life, elongation involves aminoacyl-tRNA delivered to the A site by GTP-binding elongation factors (EF-Tu in bacteria, eEF1A in eukaryotes). The core ribosomal mechanics of codon recognition and peptide-bond formation are evolutionarily conserved. Therefore, this is not fundamentally different.	
D) Stop codons are recognized by release factors that mimic tRNA — TRUE in both systems Both systems use protein release factors that structurally and functionally mimic tRNA to enter the A site and trigger hydrolysis of the nascent polypeptide. • Prokaryotes: RF1, RF2 • Eukaryotes: eRF1 This conserved mechanism is similar, not different. Pro- and eukaryotes 1. Genome Structure Eukaryotes • Linear chromosomes • Histones + chromatin • Many origins of replication • Introns present • Telomeres + telomerase Prokaryotes • Single circular chromosome	

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• No histones (except some archaea) • One origin of replication • Few/no introns • No telomeres	
2. Transcription Eukaryotes • Occurs in nucleus • Requires many general transcription factors (TFIID, TFIIF, etc.) • Promoters = TATA box, Inr, DPE • Strong chromatin regulation • Transcription and translation are separate Prokaryotes • Occurs in cytoplasm • Promoters = -10 (Pribnow box) and -35 sequences • Sigma factors initiate transcription • Minimal chromatin regulation • Transcription and translation are coupled	
3. RNA Processing Eukaryotes • mRNA is capped, polyadenylated, spliced • Alternative splicing common Prokaryotes • mRNA not capped or polyadenylated • Rare or no introns → no splicing	
4. Translation Eukaryotes • Ribosome binds 5' cap and scans for AUG • 80S ribosome (40S + 60S) • mRNAs usually monocistronic Prokaryotes • Ribosome binds Shine-Dalgarno sequence (internal initiation) • 70S ribosome (30S + 50S) • Often polycistronic mRNAs	
5. DNA Replication Eukaryotes • Multiple origins • Many DNA polymerases (α, δ, ϵ, etc.) • Must disassemble/reassemble nucleosomes • Telomerase required Prokaryotes • Single origin • Fewer polymerases (Pol III main replicative enzyme) • No telomerase required	

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6. Epigenetic Regulation Eukaryotes • Extensive: DNA methylation, histone modifications, chromatin remodeling Prokaryotes • Minimal epigenetics (e.g., Dam methylation mainly for repair/recognition)	
7. Gene Organization Eukaryotes • Genes separated • Enhancers can be far away • Alternative splicing increases diversity Prokaryotes • Operons allow coordinated expression • Rare noncoding regulatory complexity	
8. Protein Processing Eukaryotes • Heavy post-translational modifications (phosphorylation, glycosylation) • ER–Golgi trafficking Prokaryotes • Simpler modifications • No membrane-bound organelles	
★ Ultra-Condensed – main differences pro- and eukaryotes Eukaryotes: Compartmentalized → nucleus, chromatin, splicing, 5' cap + poly-A tail, scanning initiation, monocistronic mRNA, linear chromosomes with telomeres, complex regulation. Prokaryotes: Streamlined → no nucleus, no chromatin, no splicing, SD initiation, coupled transcription/translation, polycistronic mRNA, circular chromosome, operons.	
13. Which statement regarding chromosome remodeling is false?	
A) Remodeling complexes are recruited to promoters by sequence-specific activators. B) Remodeling complexes require energy in the form of ATP to modify chromatin structure C) Activation of gene expression is associated with a denser chromatin structure D) Boundary elements block the spreading of chromatin modifications E) DNA replication offer a unique possibility to remodel the chromatin structure Correct Answer (the FALSE statement): C) Activation of gene expression is associated with a denser chromatin structure This statement is false because gene activation requires open, less compact chromatin, often referred to as euchromatin. When chromatin becomes more relaxed through histone acetylation, nucleosome repositioning, or ATP-dependent remodeling, transcription factors and RNA polymerase gain access to promoters and enhancers. In contrast, dense, compact chromatin—known as heterochromatin—is	

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associated with transcriptional repression because the tightly packed nucleosomes physically obstruct the transcription machinery. Thus, the idea that gene activation correlates with denser chromatin contradicts core molecular principles, making option C the false statement.	
A) Remodeling complexes are recruited to promoters by sequence-specific activators — TRUE Chromatin remodeling complexes do not bind DNA with sequence specificity; instead, they are recruited by transcriptional activators that recognize and bind specific DNA sequences at promoters or enhancers. These activators then attract remodeling complexes, such as the SWI/SNF family, to reposition or destabilize nucleosomes, thereby promoting transcription. Because this recruitment mechanism is fundamental to regulated gene expression, option A correctly describes chromatin biology.	
B) Remodeling complexes require energy in the form of ATP to modify chromatin structure — TRUE Chromatin remodelers function as ATP-dependent molecular machines. They use the energy released from ATP hydrolysis to disrupt or reorganize histone–DNA interactions, enabling nucleosome sliding, eviction, or histone variant exchange. This ATP requirement is a defining characteristic of remodeling complexes across eukaryotes, so option B accurately reflects their mode of action.	
D) Boundary elements block the spreading of chromatin modifications — TRUE Boundary elements, also known as insulators, play a crucial role in maintaining proper chromatin domain organization. They prevent the uncontrolled spread of chromatin states such as heterochromatin, which would otherwise propagate and silence neighboring genes. Proteins like CTCF bind to boundary elements and create physical and functional barriers that preserve distinct transcriptional environments. Because this is well-established, option D is true.	
E) DNA replication offers a unique possibility to remodel the chromatin structure — TRUE During DNA replication, nucleosomes must be disassembled ahead of the replication fork and reassembled behind it. This temporary removal of histones provides an opportunity to incorporate newly synthesized histones, adjust post-translational modifications, or alter nucleosome spacing. Replication therefore creates a natural window for large-scale chromatin remodeling, making option E a correct statement about chromatin dynamics.	
14. Which of the following is NOT an example of epigenetic heritage?	
A) Stress induces gene expression B) DNA methylation	

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C) Histone modification D) X chromosome inactivation E) Prions Correct Answer: A) Stress induces gene expression This statement is not an example of epigenetic heritage because stress-induced gene expression represents a transient physiological response, not a heritable change in gene expression state. Epigenetic heritage refers to stable modifications that persist through cell division (and in some cases across generations) without altering the DNA sequence. Stress can trigger signaling pathways that temporarily activate or repress genes, but once the stimulus is removed, these changes generally do not persist, do not modify chromatin in a heritable manner, and do not qualify as epigenetic memory. Therefore, option A is not an example of epigenetic inheritance.	
B) DNA methylation — A TRUE example of epigenetic heritage DNA methylation, particularly at cytosine residues in CpG dinucleotides, is a classical epigenetic mechanism. Methylation patterns can be faithfully propagated during DNA replication via maintenance methyltransferases, allowing daughter cells to inherit the same transcriptional states as the parent cell. These patterns influence gene silencing, genomic imprinting, X-chromosome inactivation, and stable developmental programs. Thus, DNA methylation exemplifies true epigenetic inheritance.	
C) Histone modification — A TRUE example of epigenetic heritage Histones carry post-translational modifications such as methylation, acetylation, phosphorylation, and ubiquitination, all of which influence chromatin structure and gene accessibility. During DNA replication, old histones with their modifications are distributed to daughter strands, guiding the modification of newly synthesized histones. In this way, histone marks can be transmitted across cell divisions, preserving transcriptional states. Therefore, histone modification clearly represents epigenetic inheritance.	
D) X chromosome inactivation — A TRUE example of epigenetic heritage In female mammals, one X chromosome is randomly inactivated early in development through a combination of DNA methylation, histone modifications, and the action of the noncoding RNA Xist. Once inactivated, the same X chromosome remains silent in all daughter cells, making this a striking and well-characterized case of stable, heritable epigenetic regulation. Thus, option D is correctly described as epigenetic heritage.	
E) Prions — A TRUE (non-nuclear) example of epigenetic heritage	

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Prions represent an unconventional yet valid form of epigenetic inheritance in which the conformation of a protein is transmitted independently of DNA sequence. Misfolded prion proteins induce the same conformation in normal proteins, perpetuating the altered state across cell generations. This heritable transmission of phenotype without changes in nucleic acid sequences fits the broad definition of epigenetic inheritance, making prions a correct example.	
15. Which eukaryotic protein serves a similar function as the prokaryotic protein Hfq?	
A) Histone acetyltransferase B) Histones C) Dicer D) Drosha E) Argonaute Correct Answer: E) Argonaute The prokaryotic protein Hfq functions as an RNA chaperone that facilitates interactions between small regulatory RNAs (sRNAs) and their target mRNAs, stabilizing these RNAs and promoting post-transcriptional regulation. In eukaryotes, the closest functional equivalent is Argonaute, the central protein of the RNA-induced silencing complex (RISC). Argonaute binds small RNAs such as miRNAs and siRNAs and uses them as guides to identify complementary mRNA targets. Like Hfq, Argonaute mediates sequence-specific post-transcriptional regulation, including translational repression, mRNA cleavage, and mRNA destabilization. Although the molecular machinery in eukaryotes is more complex, Argonaute is the protein whose function—facilitating small RNA-guided regulation of mRNA—most closely mirrors the regulatory outcomes produced by Hfq in bacteria. Therefore, E is the correct answer.	
A) Histone acetyltransferase — Incorrect Histone acetyltransferases modify histone proteins by adding acetyl groups, leading to chromatin loosening and increased transcriptional activity. This role is entirely unrelated to RNA chaperoning or small RNA-guided regulation. Because acetyltransferases act at the level of chromatin remodeling rather than RNA-RNA interactions, they share no functional similarity with Hfq.	
B) Histones — Incorrect Histones package DNA into nucleosomes and regulate chromatin structure. They do not bind small regulatory RNAs and do not participate in post-transcriptional gene regulation. Since histones act on DNA rather than RNA and have no chaperone activity for RNA pairing, they cannot be considered analogous to Hfq in any meaningful way.	

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C) Dicer — Incorrect Dicer processes double-stranded RNA precursors into miRNAs and siRNAs, which are then loaded into Argonaute. Although Dicer participates in RNA-based gene regulation, its role is restricted to cleavage and maturation of small RNAs, not to mediating interactions between small RNAs and target mRNAs. Thus, Dicer does not function in the same manner as Hfq.	
D) Drosha — Incorrect Drosha, like Dicer, acts upstream in the miRNA maturation pathway by cleaving primary miRNA transcripts in the nucleus. Its function is enzymatic and limited to miRNA processing, not to facilitating base pairing between regulatory RNAs and mRNAs. Drosha therefore cannot be considered functionally analogous to Hfq.	
16. Which statement addressing antisense RNA is false?	
A) Antisense RNA may be fully complementary to its target B) Antisense RNA may initiate replication C) Antisense RNA may be partially complementary to its target D) Antisense RNA may act by sequestering translation initiation signals E) Antisense RNA may confer enhanced mRNA degradation Correct Answer (the FALSE statement): B) Antisense RNA may initiate replication This statement is false because antisense RNA does not initiate DNA or RNA replication. Antisense RNAs function primarily as regulators of gene expression, acting through base pairing with complementary sequences in target mRNAs. Their roles include blocking translation, promoting degradation, or altering RNA stability or processing — but they do not serve as primers or initiators for replication machinery. Replication initiation requires specialized DNA or RNA primers synthesized by dedicated enzymes such as primase, not regulatory antisense transcripts. Therefore, option B is the incorrect statement.	
A) Antisense RNA may be fully complementary to its target — TRUE Many antisense RNAs, especially in prokaryotes and synthetic antisense technologies, base-pair with mRNA targets through perfect complementarity, allowing highly specific regulatory control. Complete base pairing can block ribosome binding or recruit RNases, demonstrating that this is a valid and common mechanism.	
C) Antisense RNA may be partially complementary to its target — TRUE In eukaryotes, microRNAs (miRNAs) typically pair with their targets through imperfect complementarity, especially within the seed region. This partial matching can repress translation or destabilize the mRNA without requiring full base pairing.	

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Thus, antisense RNA does not need perfect complementarity to exert regulatory effects.	
D) Antisense RNA may act by sequestering translation initiation signals — TRUE Antisense RNAs can base-pair with regions near the ribosome-binding site (Shine-Dalgarno sequence in bacteria, or sequences around the start codon in eukaryotes), thereby blocking access of the ribosome and preventing translation. This mechanism is well established in bacterial regulatory sRNAs and in synthetic antisense approaches.	
E) Antisense RNA may confer enhanced mRNA degradation — TRUE Antisense RNA pairing can recruit RNases or lead to structural changes that make mRNAs more accessible to degradation pathways. In bacteria, Hfq-associated sRNAs often promote RNase E-mediated degradation; in eukaryotes, miRNA-RISC complexes trigger deadenylation and decay. Thus, antisense RNA-mediated mRNA degradation is a widespread regulatory mechanism.	
17. Which of the following are the basic components of a virus?	
A) ribosomes and nucleic acid B) plastids and proteins C) nucleic acid and cytoplasm D) proteins and ribosomes E) proteins and nucleic acid Correct Answer: E) proteins and nucleic acid A virus is fundamentally composed of only two essential components: its nucleic acid genome (either DNA or RNA, single- or double-stranded) and a surrounding protein coat, known as the capsid. The capsid protects the viral genetic material and enables attachment and entry into host cells. Some viruses also have a lipid envelope derived from host membranes, but this is optional and not considered a basic universal component. The defining minimal structure of all viruses is therefore nucleic acid + protein, making option E the correct answer.	
A) ribosomes and nucleic acid — Incorrect Viruses do not contain ribosomes. This absence is central to their biology: viruses cannot translate their own proteins and must rely entirely on the host's ribosomes for protein synthesis. If viruses contained ribosomes, they would resemble self-sufficient cells rather than obligate intracellular parasites. Thus, this option incorrectly includes a cellular structure absent in all viruses.	
B) plastids and proteins — Incorrect	

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Plastids (e.g., chloroplasts) are organelles found only in plants and some protists, never in viruses. Viruses lack all organelles and any form of metabolic machinery. While proteins are indeed part of viral structure, plastids have no relevance to viral composition, making this statement incorrect.	
C) nucleic acid and cytoplasm — Incorrect Although viruses contain nucleic acid, they do not contain cytoplasm. Cytoplasm is a characteristic feature of living cells, containing metabolic enzymes, ribosomes, and structural components. A virus, by contrast, is essentially a genome packaged inside protein (and sometimes lipid) with no cytoplasmic matrix of its own. Thus, this option incorrectly assigns a cellular feature to viruses.	
D) proteins and ribosomes — Incorrect Viruses contain proteins (capsid proteins, envelope proteins), but they never contain ribosomes. Because ribosomes are essential for translation, their absence is part of what makes viruses dependent on host cells. Therefore, pairing viral proteins with ribosomes is biologically inaccurate.	
18. The envelope of an enveloped virus is derived from the host's	
A) nucleic acid B) membrane structure C) cytoplasm D) genome E) ribosome Correct Answer: B) membrane structure The envelope of an enveloped virus is derived from the host cell's membrane structures, typically the plasma membrane, endoplasmic reticulum (ER), or Golgi apparatus. During viral budding, the virus acquires a lipid bilayer from these host membranes, embedding its own viral glycoproteins into the host-derived lipid envelope. This envelope helps the virus evade the immune system and facilitates entry into new host cells through membrane fusion. Because the envelope is always a host-derived lipid membrane, option B is the correct answer.	
A) nucleic acid — Incorrect Viral envelopes are not composed of nucleic acid, nor are they derived from host nucleic acids. DNA and RNA are the genetic material of the virus but play no structural role in forming the lipid envelope. Viral envelopes are made of lipids and proteins, not nucleic acids, which makes this option incorrect.	
C) cytoplasm — Incorrect Although viruses replicate inside the host cytoplasm (in many cases), the cytoplasm does not give rise to the viral envelope. The cytoplasm contains proteins, metabolites,	

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and organelles, but it is not a membrane system. The viral envelope must come from a lipid bilayer , which the cytoplasm does not supply. Therefore, this choice is incorrect.	
D) genome — Incorrect A viral genome, whether DNA or RNA, encodes viral proteins and regulatory elements but does not provide structural components of the envelope. The genome remains inside the virus particle, protected by the capsid and envelope, but it does not contribute lipids or membrane material. Thus, this cannot be the origin of the envelope.	
E) ribosome — Incorrect Ribosomes synthesize proteins but are not membrane structures and are never incorporated into viral envelopes. In fact, the absence of ribosomes is a key distinguishing feature of viruses, underscoring their reliance on host translation machinery. Since ribosomes cannot form lipid bilayers, they are irrelevant to envelope formation.	
19. Which of the following is a change in the sequence that leads to formation of a stop codon?	
A) missense mutation B) nonsense mutation C) silent mutation D) genetic drift E) gene duplication Correct Answer: B) nonsense mutation A nonsense mutation is a point mutation in which a codon that previously encoded an amino acid is changed into a stop codon (UAA, UAG, or UGA). This premature stop halts translation and produces a truncated protein, which is often nonfunctional or rapidly degraded via nonsense-mediated decay. Because a nonsense mutation <i>specifically</i> converts a sense codon into a termination codon, it is the only option that correctly describes the type of sequence change leading to stop codon formation, making B the correct answer.	
A) missense mutation — Incorrect A missense mutation is a single nucleotide change that results in the substitution of one amino acid for another within the protein sequence. The codon continues to encode an amino acid, and no stop codon is created. Missense mutations may alter protein function, but they do not introduce premature termination, so this option is incorrect in the context of stop codon formation.	
C) silent mutation — Incorrect	

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A silent mutation is a change in the DNA sequence that does not alter the encoded amino acid, typically due to the redundancy of the genetic code (codon degeneracy). Because the amino acid remains the same, the mutation by definition does not produce a stop codon and has no effect on protein length. Therefore, this cannot be the type of mutation that generates a stop codon.	
D) genetic drift — Incorrect Genetic drift refers to random changes in allele frequencies within a population over time. It is an evolutionary process, not a specific type of DNA sequence change. While drift may influence the prevalence of mutations (including nonsense mutations), it does not itself describe the molecular event that introduces a stop codon. Thus, this option is conceptually unrelated to the question.	
E) gene duplication — Incorrect Gene duplication is a large-scale genomic event resulting in the copy of an entire gene or genomic region. This process increases gene dosage and contributes to evolutionary innovation but does not inherently generate stop codons. Duplication affects genome structure, not individual codons, making this option incorrect.	
20. The core genome of a species is the genomic content that is	
A) Common to all strains of the same species B) Present in only one strain of the species C) Hypothetical or uncharacterized genome content of a species D) Shared with all other prokaryotes E) Present on plasmids Correct Answer: A) Common to all strains of the same species The core genome of a species refers to the set of genes that are shared by all strains within that species. These genes typically encode essential cellular functions such as replication, transcription, translation, and primary metabolism. Because these functions are required for the fundamental biology and survival of the species, they are conserved across every strain, regardless of ecological niche or horizontal gene transfer events. In microbial genomics, the core genome contrasts with the accessory (or dispensable) genome, which varies among strains and contributes to differences in virulence, antibiotic resistance, environmental adaptation, and metabolic diversity. Thus, the core genome is defined specifically as the common genomic content present in all members of a species, making option A the correct answer.	
B) Present in only one strain of the species — Incorrect Genes present in only one strain are not part of the core genome but instead belong to the accessory genome or strain-specific genome. These unique genes often arise from horizontal gene transfer, mobile genetic elements, or niche-specific adaptations.	

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Because the definition of the core genome requires universality across all strains, genes exclusive to a single strain cannot be included, making this option incorrect.	
C) Hypothetical or uncharacterized genome content of a species — Incorrect Hypothetical or uncharacterized genes refer to genes whose functions are unknown, but their status has no direct relationship to whether they belong to the core or accessory genome. Functionally uncharacterized genes may be core if they are present in all strains, or they may be accessory if they occur in only some. The term “hypothetical” therefore describes annotation quality, not genomic distribution, making this option unrelated to the definition of a core genome.	
D) Shared with all other prokaryotes — Incorrect Genes shared across all prokaryotes represent universal genes, often involved in the most fundamental cellular processes such as ribosomal function or fundamental metabolic pathways. These genes transcend species boundaries and are conserved across entire domains of life, not restricted to a single species. Although universal genes may appear in a species’ core genome, the core genome is species-specific, not universal across prokaryotes. Therefore, this option is incorrect.	
E) Present on plasmids — Incorrect Plasmid genes are typically accessory genes, not core genes. Because plasmids can be gained or lost and vary widely between strains, they do not satisfy the requirement of being present in all strains of a species. Plasmids frequently carry adaptive traits such as antibiotic resistance or virulence factors, which are not universally required. As a result, plasmid content cannot define the core genome, making option E incorrect.	

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Assignment 12. Virus

Task set A

Task set A contains 10 multiple choice questions. The correct answer should be indicated by a letter in the cell next to the possible answers.

	correct answer
1) Rolling circle replication generates: A) a single-stranded linear single copy. B) a double-stranded linear single copy. C) a single-stranded circular single copy. D) a multimeric single-stranded “tail.”	
Correct answer: D) a multimeric single-stranded “tail.” Rolling-circle replication is a mechanism used by many plasmids, bacteriophages (e.g., ϕX174), and some viral genomes. It begins when an endonuclease introduces a nick in one strand of a circular DNA molecule. The 3'-OH produced by that nick serves as a primer for DNA polymerase, which extends the strand while displacing the old complementary strand. Because DNA synthesis proceeds continuously around the circle, the polymerase can pass the original nick site repeatedly, each time displacing more of the original strand. This results in the production of a long, continuous, single-stranded DNA tail composed of multiple genome-length units linked end-to-end (a multimeric ssDNA concatemer)—exactly what option D describes.	
Why the other options are incorrect A) “a single-stranded linear single copy” — Incorrect. Although rolling-circle replication <i>does</i> generate linear single-stranded DNA, it is not limited to a single genome-length unit. Polymerase continues synthesis around the template multiple times unless terminated, producing multimeric (concatemeric) ssDNA rather than a single-copy linear strand. B) “a double-stranded linear single copy” — Incorrect. The displaced product of rolling-circle replication is single-stranded, not double-stranded. A double-stranded linear product may arise <i>later</i> if the ssDNA is used as a template for complementary-strand synthesis, but this is not the immediate product of the rolling-circle mechanism. C) “a single-stranded circular single copy” — Incorrect. The displaced strand is released as a linear molecule, not a circular one. Circularization typically requires additional enzymes and is not the direct result of the replication fork moving around the circle.	

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2) In viruses, genetic information flows generally from _____ to _____. A) RNA / virion B) capsid / virion C) DNA / protein D) nucleic acid / protein Correct answer: D) nucleic acid → protein In viruses, the most fundamental and universally conserved flow of genetic information is from the viral nucleic acid—whether DNA or RNA—to the viral proteins that make up the structural components (such as the capsid) and the enzymes required for replication. Regardless of genome type or replication strategy, a virus must express its genes to produce proteins, and these proteins ultimately assemble into new virions. Thus, the most general and theoretically correct statement of information flow in viruses is from nucleic acid to protein.	
Why the other options are incorrect A) RNA → virion — Incorrect. While many viruses use RNA genomes, not all do; DNA viruses also exist. Moreover, information does not flow directly from RNA to a <i>virion</i>. A virion is an assembled particle made of proteins and nucleic acids. The information flows first to proteins, which are then assembled into virions. Thus this option is both incomplete and conceptually misphrased. B) capsid → virion — Incorrect. The capsid is a protein shell, not a source of genetic information. It neither stores nor transmits genetic instructions. Instead, it is the <i>product</i> of expressed viral genes and forms part of the virion. Information flows toward capsid proteins during viral replication, not from them. C) DNA → protein — Incorrect as a “general” statement. Although DNA viruses follow a DNA → RNA → protein flow, many viruses—including positive-sense RNA viruses, negative-sense RNA viruses, and retroviruses—do not have DNA at all. Because the question asks what is generally true for viruses as a whole, restricting the genome to DNA is incorrect. Viral information originates from whatever nucleic acid the virus possesses, be it RNA or DNA.	
3) What would be the consequence of deleting the late T4 genes? A) T4 mRNA would not be produced. B) ATP would not be produced and the T4 genome would not be packaged into the capsid. C) The T4 genome would not be copied. D) T4 capsid proteins would not be made. Correct answer: D) T4 capsid proteins would not be made.	

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In bacteriophage T4, gene expression is temporally regulated in three waves: early, middle, and late. The late genes encode the structural components of the virion—most notably capsid proteins, tail fibers, baseplates, and the proteins involved in assembly and packaging of the replicated DNA. If these late genes are deleted, the virus will still replicate its DNA and synthesize early and middle gene products, but it will be unable to produce the structural proteins required to assemble new phage particles. Consequently, no mature virions can form. This directly corresponds to option D.

Why the other options are incorrect

A) “T4 mRNA would not be produced.” — Incorrect.

mRNA production occurs from early and middle genes before late gene transcription. Deleting late genes does not impair the transcription of early or middle genes; the cell and phage transcription machinery would still produce mRNA from the intact earlier gene sets. Only the mRNA corresponding to the late genes would be missing, not *all* T4 mRNA.

B) “ATP would not be produced and the T4 genome would not be packaged into the capsid.” — Incorrect.

ATP production is a host cell process, not encoded by T4 late genes. Deleting late genes would not halt ATP synthesis in the bacterium. Although DNA packaging would indeed fail (because packaging proteins are late-gene products), this answer incorrectly claims that ATP production itself would cease.

C) “The T4 genome would not be copied.” — Incorrect.

Genome replication depends on early and middle gene products, such as T4 DNA polymerase and replication-associated proteins. These remain functional even if late genes are deleted. Therefore, the viral genome would still be synthesized successfully.

4) The discovery of retroviruses changed our understanding of

A) infectious particles.

B) protein synthesis.

C) the flow of genetic information.

D) gene structure and organization.

Correct answer: C) the flow of genetic information.

The discovery of retroviruses fundamentally altered molecular biology’s central dogma. Until retroviruses were studied, the accepted flow of genetic information was DNA → RNA → protein. Retroviruses, however, possess RNA genomes and use a unique enzyme, reverse transcriptase, to synthesize DNA from an RNA template. This reverse flow—RNA → DNA—was previously unknown in nature. Its discovery forced a revision of the central dogma to accommodate reverse transcription and demonstrated that genetic information can move in more directions than originally believed. Thus, the key conceptual impact of retroviruses was on our understanding of the flow of genetic information.

Why the other options are incorrect A) “infectious particles” — Incorrect. Retroviruses did not change our basic understanding of what constitutes an infectious particle. Viruses had long been known to be infectious agents composed of nucleic acid and protein. Retroviruses fit into this established framework; they merely use an unusual replication strategy rather than redefining what an infectious particle is. B) “protein synthesis” — Incorrect. Retroviruses do not contradict or alter the mechanism of protein synthesis. Their mRNAs are translated by the host cell’s standard translational machinery, following the same rules of codons, ribosomes, and tRNAs as all other organisms and viruses. Nothing about the discovery of retroviruses required a reinterpretation of how proteins are made. D) “gene structure and organization” — Incorrect. Although retroviral proviruses integrate into host genomes, this did not fundamentally change the general concept of gene structure. The existence of introns, exons, operons, and other structural features was already known from cellular genes. Retroviruses added interesting examples but did not redefine gene architecture in the same transformative way they altered the central dogma.	
5) Which type of viruses integrate their genetic material within the host genome?	
A) Retroviruses B) Megaviruses C) Adenoviruses D) bacteriophages and retroviruses Correct answer: D) bacteriophages and retroviruses. Integration of viral genetic material into the host genome is a hallmark of two major groups: 1. Retroviruses, which <i>obligately</i> integrate a DNA copy of their RNA genome into the host chromosome via the enzyme integrase. This integrated DNA (the provirus) becomes a permanent part of the host genome. 2. Temperate bacteriophages (such as λ phage), which can enter the lysogenic cycle, during which the viral genome integrates into the bacterial chromosome as a prophage, remaining latent until induction. Because both retroviruses and certain bacteriophages are characterized by genome integration, option D is the most complete and correct answer.	
Why the other options are incorrect A) Retroviruses — Incomplete. Although retroviruses <i>do</i> integrate into the host genome, they are not the only viruses that do so. Temperate bacteriophages also integrate their genomes as part of their life cycle. Therefore, this option is correct but incomplete and thus not the best answer.	

B) Megaviruses — Incorrect. Megaviruses (large DNA viruses such as Mimivirus) replicate in the cytoplasm and do <i>not</i> integrate their genomes into the host chromosome. No integrase-like mechanism is part of their life cycle. C) Adenoviruses — Incorrect. Adenoviruses maintain their DNA as episomal, non-integrated units in the nucleus. They replicate independently and do not integrate their DNA into the host genome as part of their normal biology.	
6) Viral replication is	
A) independent of the host cell's DNA but dependent on the host cell's enzymes and metabolism. B) dependent on the host cell's DNA, RNA, enzymes, and metabolism. C) independent of both the host cell's DNA and the host cell's enzymes and metabolism. D) dependent on the host cell's DNA and RNA. Correct answer: A) independent of the host cell's DNA but dependent on the host cell's enzymes and metabolism. Viral replication fundamentally relies on host-cell machinery. Viruses lack the ribosomes, metabolic pathways, and—in many cases—the polymerases required for genome replication and protein synthesis. Thus, they must use the host’s enzymes, energy sources, and metabolic infrastructure to reproduce. However, viral replication is not dependent on the host’s DNA itself. A virus does not need the host genome to be replicated or expressed; it needs only the <i>enzymatic machinery</i> that the host cell provides. Many viruses replicate efficiently in enucleated cells (if the necessary cytoplasmic machinery remains) or possess their own genome-processing enzymes, but none can replicate without host metabolism. This makes option A theoretically precise.	
Why the other options are incorrect B) “dependent on the host cell's DNA, RNA, enzymes, and metabolism” — Incorrect. Viruses do not require the host’s DNA or RNA sequences to replicate; they only need the host’s transcriptional and translational machinery, and sometimes not even that (e.g., large DNA viruses encode their own polymerases). The host DNA itself plays no direct role in viral replication. Thus, this option overstates the dependency. C) “independent of both the host cell's DNA and the host cell's enzymes and metabolism” — Incorrect. This contradicts the most basic principle of virology: viruses cannot replicate independently. Without access to host enzymes, ribosomes, and metabolic energy, viral replication is impossible. No known virus is metabolically autonomous.	

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D) “dependent on the host cell's DNA and RNA” — Incorrect. Again, viruses do not depend on the <i>host's</i> nucleic acids. They carry their own genomes, and these alone guide replication and protein synthesis. What they require is the host's enzymatic and metabolic capabilities, not the host's genetic material.	
7) A virus that kills its host is said to be A) lytic or virulent. B) lysogenic. C) temperate. D) virulent or lysogenic, but not temperate. Correct answer: A) lytic or virulent. A virus that kills its host engages in a lytic cycle, in which infection leads to rapid replication of viral particles, destruction of the host cell membrane, and eventual lysis of the cell. Viruses that exclusively follow this lethal strategy are termed virulent. Thus, a virus that kills its host is correctly described as lytic or virulent, making option A the accurate choice.	
Why the other options are incorrect B) lysogenic — Incorrect. A lysogenic virus <i>does not</i> kill its host. Instead, it integrates its genome into the host's chromosome, forming a prophage, and remains dormant. The host cell survives and continues dividing, carrying the viral genome along. Only upon induction does a lysogenic virus switch to the lytic cycle, at which point it becomes destructive. Therefore, lysogenic is the opposite of what the question describes. C) temperate — Incorrect. Temperate viruses are capable of both lysogenic and lytic cycles. They do <i>not</i> necessarily kill the host. In fact, their defining feature is their ability to exist harmlessly within the host genome in a lysogenic state. They <i>may</i> kill the host if induced into the lytic cycle, but they are not defined by lethality. Thus, “temperate” does not correctly describe a virus that kills its host. D) virulent or lysogenic, but not temperate — Incorrect. This option contradicts fundamental virological definitions. • “Virulent” is correct for a host-killing virus. • “Lysogenic” is incorrect because lysogenic viruses do not kill their hosts. • “Not temperate” is irrelevant because temperate viruses can be lytic but are not defined by killing. Thus the statement is internally inconsistent and biologically incorrect.	
8) Restriction is	
A) a general host mechanism to prevent the invasion of foreign nucleic acid. B) a general host mechanism to degrade viral proteins. C) the viral process whereby a host's DNA ceases normal functioning. D) the viral process whereby the virus prevents other viruses from entering the cell.	

Correct answer: A) a general host mechanism to prevent the invasion of foreign nucleic acid.

“Restriction” refers to the defense strategy employed by bacteria in which restriction endonucleases recognize and cleave foreign DNA—most commonly invading bacteriophage genomes. This system discriminates between self and non-self: host DNA is typically protected by methylation, while unmethylated viral DNA is cut. Thus, restriction serves as a host-driven, sequence-specific mechanism aimed at preventing successful infection by degrading the incoming foreign nucleic acid. This makes option A the theoretically correct and biologically precise choice.

Why the other options are incorrect

B) “a general host mechanism to degrade viral proteins” — Incorrect.

Restriction systems target DNA, not proteins. Although cells do have proteolytic pathways, these are not referred to as “restriction,” nor are they the primary antiviral defense in bacteria. Restriction endonucleases act exclusively on nucleic acids.

C) “the viral process whereby a host's DNA ceases normal functioning” — Incorrect.

Restriction is not a viral strategy. It is a host defense. Additionally, it does not interfere with host DNA function; instead, it protects host DNA by leaving it intact while degrading foreign DNA.

D) “the viral process whereby the virus prevents other viruses from entering the cell” — Incorrect.

This describes superinfection exclusion, a viral mechanism, not restriction. Restriction endonucleases do not regulate viral entry; they act *after* foreign DNA enters the cell, cutting it internally. Thus, the concept is unrelated to restriction.

9) Reverse transcriptase is a(n)

A) RNA-dependent DNA polymerase.

B) DNA-dependent RNA polymerase.

C) DNA-dependent DNA polymerase.

D) RNA-dependent RNA polymerase.

Correct answer: A) RNA-dependent DNA polymerase.

Reverse transcriptase is the hallmark enzyme of retroviruses and retrotransposons. Its defining activity is the ability to synthesize DNA using an RNA template—a reversal of the usual information flow described in the central dogma. This capability allows an RNA genome to be converted into complementary DNA (cDNA), which can then integrate into the host genome. Because the enzyme requires RNA as the template and produces DNA as the product, it is precisely classified as an RNA-dependent DNA polymerase, making option A the correct and theoretically accurate choice.

Why the other options are incorrect

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B) DNA-dependent RNA polymerase — Incorrect.

This describes RNA polymerase, the enzyme responsible for transcription in all cells. It uses DNA as the template to produce RNA, which is the opposite direction from reverse transcription.

C) DNA-dependent DNA polymerase — Incorrect.

This refers to classical DNA polymerases involved in genome replication in cells and many DNA viruses. These enzymes copy DNA from DNA and play no role in reverse transcription.

D) RNA-dependent RNA polymerase — Incorrect.

This enzyme is employed by RNA viruses (e.g., influenza, coronaviruses) to replicate their RNA genomes directly from RNA templates. Although it also uses RNA as a template, it produces RNA, not DNA, and is therefore fundamentally distinct from reverse transcriptase.

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Task set B

Task set B contains 3 essay questions (4 in the final exam). The answer should be written using the space allocated below each question.

1) The genome of an unknown virus was found to contain 40 mol% adenine, 20 mol% thymine, 15 mol% guanine and 25 mol% cytosine. What can you terminate about the structure of this virus' genetic material?

The base composition of the unknown viral genome—40 mol% adenine, 20 mol% thymine, 15 mol% guanine, and 25 mol% cytosine—allows us to draw clear conclusions about the nature of its genetic material. The presence of thymine immediately indicates that the genome is composed of DNA rather than RNA, since RNA uses uracil in place of thymine. Once established as DNA, the next question is whether it is double-stranded or single-stranded. In any double-stranded DNA molecule, Chargaff's rules apply, meaning that the proportions of adenine and thymine must be equal, as must the proportions of guanine and cytosine. The values given, however, violate both of these requirements: adenine does not equal thymine, and guanine does not equal cytosine. Such an asymmetrical base composition cannot occur in double-stranded DNA, where every base is paired with its complement. Therefore, the only consistent interpretation is that this virus contains a single-stranded DNA genome, a feature characteristic of certain virus families such as the parvoviruses.

2) Give a definition of a virulent and a temperate phage.

A virulent phage is a bacteriophage that reproduces exclusively through the lytic cycle. Upon infection, it immediately hijacks the host cell's machinery to replicate its genome and produce new phage particles, ultimately causing lysis and death of the host cell. Virulent phages do not integrate into the host chromosome and have no latent or dormant phase.

A temperate phage, in contrast, is a bacteriophage capable of choosing between two reproductive pathways. It may enter the lytic cycle, killing the host, or it may enter the lysogenic cycle, in which its genome integrates into the host's chromosome and becomes a prophage. In this lysogenic state, the viral genome is replicated passively as the host cell divides, and the cell remains alive. The prophage can later be induced to re-enter the lytic cycle under specific conditions.

In short:

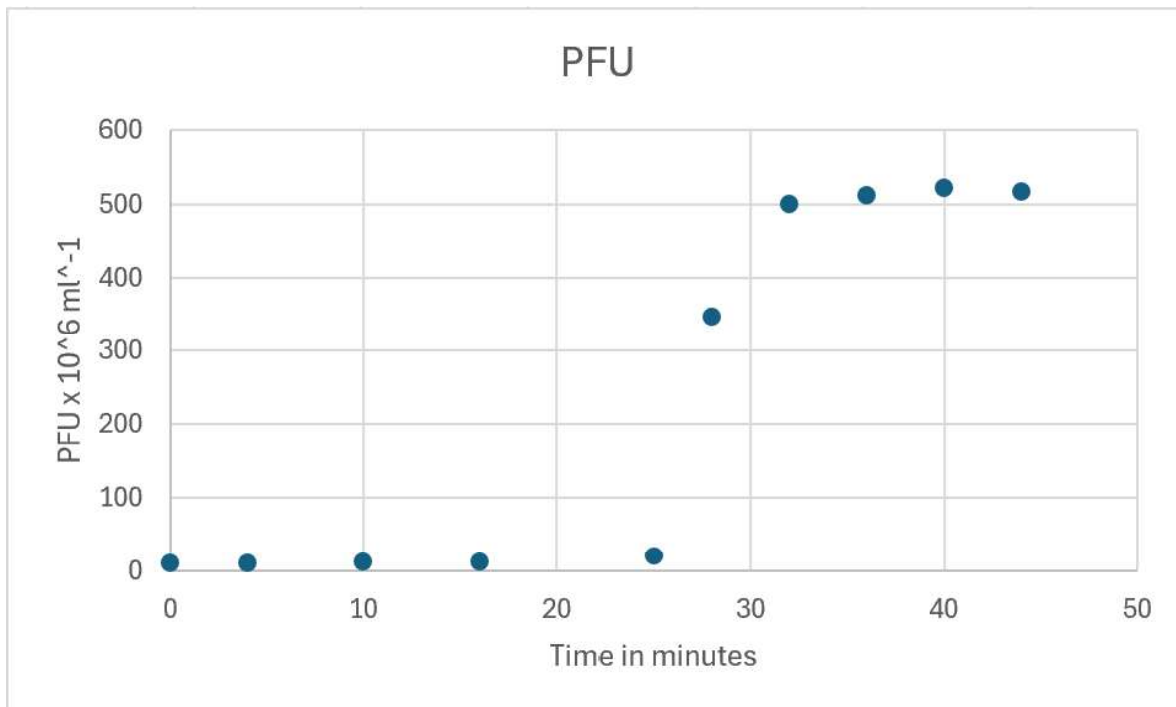
- Virulent phage: always lytic, always kills the host.
- Temperate phage: can be lytic or lysogenic, may either kill the host or persist within it.

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3) A one-step growth experiment was performed with a new phage W1971. The host cells were infected at time 0 with W1971. The number of plaque forming units per ml is arbitrarily set to 1 at the phage addition. The number of bacteria was much greater than the number of bacteriophages added, i.e. that a given bacterium is infected only by one subject. The number of plaque forming units (PFU) was determined in a plaque assay. The results are shown in the table below:

Time in minutes	PFU $\times 10^6 \text{ ml}^{-1}$
0	9
4	10
10	11
16	12
25	19
28	345
32	500
36	511
40	521
44	517

Question 1. Find the latency time and burst size measured in this experiment – eventually after plotting the data.



The latent period is the time from infection until the first detectable release of new phage i.e. up to the start of the rise. So the latency time is about 25 mins.

Then to find the burst size we need the phage mean of PFU (number of plaque forming units) before the rise and at the plateau.

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$$\text{Burst size} = \left(\frac{\text{phage at plateau}}{\text{phage before rise}} \right)$$

$$\text{Phage at plateau} = \text{mean of PFU at 32-44 min} = \frac{500+511+521+517}{4} = 512,25$$

$$\text{Phage before rise} = \text{mean of PFU at 0-16 min} = \frac{9+10+11+12}{4} = 10,5$$

$$\text{Burst size} = \frac{512,25}{10,5} \approx 49$$

So the burst size is about 49 phages per infected cell

Esperanzas answer:

The latency period is the period of absence of free bacterial phage, and throughout this period the number of plaques will be the same as the number of bacteriophages that occurred at the beginning of the trial. The latency time is 25 mins. Burst size is 48 (512/10.5). It is the number of bacteriophages that are formed (and released) from one infected bacterium

During a similar one-step growth experiment, samples are taken at specific time points for the determination of mRNA produced from different phage genes. The genes examined are A, B, C, D, E, F, G and H. As shown in Table 1, the synthesis of mRNA from the various phage genes showed a temporal dependency.

Table 1. Infection with phage W1971

Time after phage addition	Production of mRNA ^a							
	<i>genA</i>	<i>genB</i>	<i>genC</i>	<i>genD</i>	<i>genE</i>	<i>genF</i>	<i>genG</i>	<i>genH</i>
0 min	-	-	-	-	-	-	-	-
10 min	+	+	+	+	-	-	-	-
20 min	-	-	-	-	+	+	+	+
30 min	-	-	-	-	+	+	+	+

a) + means the presence of specific mRNA, - means that specific mRNA is not present.

Question 2. Describe the time course of mRNA synthesis from genes A-H. How would you classify the genes A-D and the genes E-H?

The data show that the phage genes fall into two clearly separated temporal classes of transcription. At the moment of infection (0 minutes), no phage-specific mRNA is detectable, which is expected because the phage has not yet begun to redirect the host's transcription machinery. By 10 minutes, however, mRNA from genes A, B, C, and D appears. These transcripts are present only at this early time point and disappear by 20 minutes, indicating that genes A–D are expressed immediately after infection and then rapidly shut off. Such genes typically encode regulatory and replication functions—proteins that modify the host RNA polymerase, degrade host DNA, or initiate phage genome replication. Their short-lived expression pattern is characteristic of **early genes**.

In contrast, genes E, F, G, and H show no expression at 10 minutes. Their mRNA first appears at 20 minutes and continues to be detectable at 30 minutes. This delayed and sustained pattern identifies them as **late genes**, which commonly encode structural components of the phage, such as capsid and tail proteins, as well as packaging factors required to assemble new virions. Late gene expression begins only after early genes have prepared the host for phage replication.

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Thus, the time course of mRNA synthesis reveals two transcriptional waves: an early wave involving genes A–D, followed by a late wave involving genes E–H. This temporal program reflects the typical organization of a lytic phage life cycle, in which early functions enable replication, while late functions construct new phage particles.

Esperanza's answer:

The genes A to D are expressed relatively early in the bacteriophage's life cycle and genE to genH late. They are therefore early and late expressed genes, respectively. It is also seen that the early and late genes cannot have common promoters since mRNAs from genes A-D and E-H are not present at the same time. It can also be seen that the gene expression from genA-D is repressed late in the phage's life cycle (by 30 min).

Task set C

Task set C contains a case problem with two sub-questions (four in the exam). The answers to each sub-question should be written using the space allocated below each question.

For a research project, you have contacted a Contract Research Organization (CRO) to get a llama immunized with a bacterial virulent factor that causes enterohemorrhagic diarrhea. You aim to find antibodies that can neutralize the viral factor and be used for prophylactic treatment in areas with poor water quality and food hygiene.

The library contains 1×10^9 unique antibodies in a phage M13 scaffold, and the CRO sent it in at a concentration of 1×10^{12} pfu/mL.

The first quality check in your laboratory is to multiply the library. To do so, you start a culture of *E.coli* TG1 in the lab and grow until it is in the exponential phase ($OD_{600}=0.5$).

Note:

- $OD_{600}=1$ in an *E.coli* culture is equivalent to 8×10^8 cells/ml
- Pfu = plaque forming units

1. Why do you need to grow *E.coli* TG1 to the exponential phase?

Growing *E. coli* TG1 to the exponential (log) phase is essential because this is the physiological state in which the bacteria are healthiest, metabolically most active, and most competent to support infection by filamentous phages such as M13, which carry the antibody library. In exponential phase, cells are continuously synthesizing DNA, RNA, proteins, and membrane components, which provides all of the resources required for efficient phage entry, replication, and assembly. Importantly, M13 and related filamentous phages infect only F^+ cells, using the F-pilus as their receptor. The density and activity of F-pili are significantly higher when the bacteria are in exponential growth. If the culture were in stationary phase, the cells would down-regulate metabolic activity and reduce the number of F-pili, making phage infection inefficient or even impossible. Moreover, phage display libraries rely on maintaining the full diversity of the 10^9 unique antibody variants. Efficient infection and replication require that each phage particle finds a host cell quickly and that the bacterial host is capable of robust phage production. Exponentially growing cells provide optimal conditions for this, ensuring that the entire library is amplified uniformly without losing variants.

Thus, *E. coli* must be in the exponential phase to:

- maximize F-pilus expression and therefore phage infection efficiency,
- provide strong metabolic activity for high phage output,

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- ensure uniform and faithful amplification of the phage library.

For these reasons, exponential-phase growth is a critical requirement when multiplying an M13 phage–display library.

Important to understand here:

This question tests the understanding that viruses are obligate intracellular parasites and can only replicate in metabolically active host cells. For bacteriophages such as M13, efficient infection requires that the host expresses the appropriate receptor—in this case, the F-pilus, which is produced primarily during the exponential phase of bacterial growth. Only log-phase *E. coli* provide the high metabolic activity, active DNA and protein synthesis, and membrane turnover needed to support phage entry, replication, and assembly. Therefore, to successfully amplify a phage library and maintain its diversity, the host culture must be in exponential growth where both infection efficiency and phage production are optimal.

Esperanza's answer:

M13 phage infects *E. coli* by attaching to the F pilus, which is only expressed in the exponential phase of growth

2. To ensure you are multiplying the whole diversity of the library, you want to infect with a number of phages 100x greater than the library diversity. How many mL of *E. coli* TG1 in the exponential phase are necessary to ensure a MOI=10?

To get 100x the number of phages we need to know the library diversity which we are told is 10^9 different antibodies. So we need to multiply this number by 100:

$$100 \cdot 10^9 = 10^{11} \text{ phages}$$

We want a MOI=10, which means 10 phages per bacterial cell, so the number of bacterial cells needed is:

$$N_{\text{cells}} = \frac{N_{\text{phage}}}{\text{MOI}} = \frac{10^{11}}{10} = 10^{10} \text{ cells}$$

We are also told that an *E. coli* culture at $OD_{600} = 1$ corresponds to $8 \cdot 10^8$ cells/mL and our culture is at $OD_{600} = 0.5$, so it contains

$$\text{cell concentration} = 4 \cdot 10^8 \text{ cells/mL}$$

So the volume of this culture required to give 10^{10} cells is:

$$V = \frac{N_{\text{cells}}}{\text{cell concentration}} = \frac{10^{10}}{4 \cdot 10^8} = 25 \text{ mL}$$

So we need 25 mL of *E. coli* TG1 culture in exponential phase to achieve an MOI=10.

What is an MOI?:

$$\text{MOI} = \frac{\text{number of infected phage particles}}{\text{number of host cells}} = \frac{N_{\text{phage}}}{N_{\text{cells}}}$$

So, an MOI=10 means that on average, 10 phages are added per 1 bacterial cell.

Esperanza's answer:

To represent 10 times every antibody in the library, for a library with 1×10^9 unique antibodies, we need 1×10^{10} pfu. If we want to infect *E. coli* in the exponential phase with an MOI=10, we need 10

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times more phages than cells, which means we need 1×10^{10} cells. In the exponential phase $OD_{600}=0.5$, the concentration of cells in the culture will be approximately 4×10^8 cells/ml. To have 1×10^{10} cells in the exponential phase in the culture, we need 25 mL of culture (1×10^{10} cells/ 4×10^8 cells/ml = 25 mL).

3. How many mL of the library sent by the CRO do you have to add to a 50 mL culture if you want to ensure a $MOI=10$

We are given that:

- Phage titer from CRO = $T = 10^{12}$ PFU/mL
- Culture volume = 50 mL
- Cell density ($OD_{600} = 0.5$) = $4 \cdot 10^8$ cells/mL
- Desired $MOI = 10$

First, we calculate number of cells in 50 mL:

$$N_{cells} = 50 \cdot 4 \cdot 10^8 = 2 \cdot 10^{10}$$

Next, we calculate phage needed for $MOI=10$:

$$N_{phage_needed} = MOI \cdot N_{cells} = 10 \cdot 2 \cdot 10^{10} = 2 \cdot 10^{11}$$

Then, we convert this into volume of the CRO library:

$$V_{phage} = \frac{N_{phage_needed}}{T} = \frac{2 \cdot 10^{11}}{10^{12}} = 0.2 \text{ mL}$$

So, you must add 0.2 mL of the CRO phage library to maintain an $MOI=10$ in a 50 mL exponentially growing E. Coli TG1 culture.

Esperanza's answer:

The infection will take place when the $OD_{600}=0.5$. The concentration of cells in the culture will be approximately 4×10^8 cells/ml. This means that in 50 mL of culture, there will be 2×10^{10} cells. For a $MOI=10$, I need 10 phages per cell, which are 2×10^{11} pfu. If the aliquots from the CRO are at 1×10^{12} pfu/mL, I need 0.2 mL of the stock (2×10^{11} pfu/ 1×10^{12} pfu/mL = 0.2 mL)

Assignment 11. Mobile genetic elements (like transposons)

Task set A

Task set A contains 10 multiple choice questions (20 in final exam). The correct answer should be indicated by a letter in the cell next to the possible answers.

	Correct answer
1) Conjugation mediates transfer of	
A) dsDNA from one bacterium to another B) ssDNA from one bacterium to another C) dsRNA from one bacterium to another D) ssRNA from one bacterium to another Correct answer: B) ssDNA from one bacterium to another Bacterial conjugation is a process in which a donor cell transfers genetic material to a recipient cell through a physical connection called a conjugation pilus or mating bridge. During this process, the plasmid or mobilizable DNA is not transferred as double-stranded DNA. Instead, one strand of the plasmid DNA is nicked, and a single-stranded DNA (ssDNA) molecule is actively transported into the recipient cell. Both the donor and recipient then synthesize the complementary DNA strand, restoring double-stranded plasmids in each cell. Thus, the actual transferred molecule is ssDNA, making option B correct.	
Why the other answers are wrong A) dsDNA from one bacterium to another Although both donor and recipient end up with double-stranded plasmids after replication, the DNA is not transferred in double-stranded form. The conjugation machinery is designed to move only one strand across the mating bridge. Therefore, this option is incorrect. C) dsRNA from one bacterium to another Bacteria do not use RNA as hereditary material, nor do they transfer RNA duplexes during conjugation. This option is unrelated to the biology of bacterial gene transfer and is incorrect. D) ssRNA from one bacterium to another Again, RNA is not transferred during bacterial conjugation. The mechanism specifically involves DNA, not RNA. No bacterial conjugation system transports RNA molecules as genetic material, so this answer is incorrect.	
2) An example of an episome is	
A) The telomeres B) The F-plasmid C) The centrosome	

D) The nucleosome Correct answer: B) The F-plasmid An episome is a genetic element—typically a plasmid—that can exist in two forms within a cell: 1. As an independent, self-replicating DNA molecule, and 2. Integrated into the host chromosome as part of the genomic DNA. The F-plasmid of <i>Escherichia coli</i> is the classic example of such an element. It can replicate autonomously in the cytoplasm, but it can also integrate into the bacterial chromosome through homologous recombination. When integrated, it can mediate high-frequency transfer of chromosomal genes during conjugation. This dual lifestyle defines it as an episome, making B the correct answer.	
Why the other answers are wrong A) The telomeres Telomeres are specialized DNA–protein structures at the ends of eukaryotic chromosomes. They protect chromosome ends from degradation and fusion, but they cannot exist independently of the chromosome and cannot integrate elsewhere. Therefore, telomeres are not episomes.	
C) The centrosome The centrosome is a protein-based organelle involved in microtubule organization and spindle formation during mitosis. It contains no plasmid-like genetic element and does not replicate autonomously through DNA-based mechanisms. Thus, it is not an episome.	
D) The nucleosome A nucleosome is the basic unit of chromatin structure: DNA wrapped around histone proteins. It is simply part of the chromosome's packaging system and cannot exist as an independent replicating DNA element. Therefore, it is not an episome.	
3) Which type of polymerase is required for turning RNA into DNA	
A) Reverse transcriptase B) tRNA polymerase C) Reverse DNA polymerase I D) Primase Correct answer: A) Reverse transcriptase Reverse transcriptase is the enzyme that synthesizes DNA from an RNA template. It is used by retroviruses, retrotransposons, and certain laboratory methods (such as RT-PCR) to generate complementary DNA (cDNA). The defining feature of reverse transcriptase is its ability to perform RNA-dependent DNA polymerization, meaning it can copy RNA and produce DNA. This is exactly the activity described in the question.	

Why the other answers are wrong B) tRNA polymerase There is no enzyme called “tRNA polymerase” that converts RNA into DNA. RNA polymerases transcribe DNA into RNA, not the other way around, and they do not use RNA as a template. Therefore, this option is incorrect.	
C) Reverse DNA polymerase I This enzyme does not exist. DNA polymerase I in bacteria synthesizes DNA from a DNA template, not from RNA. It cannot convert RNA into DNA, nor does it have reverse-transcriptase activity. Thus, it is not the correct answer.	
D) Primase Primase synthesizes short RNA primers during DNA replication. It creates RNA, but it does not turn RNA into DNA. It simply provides a starting point for DNA polymerases. Therefore, primase is unrelated to RNA-to-DNA conversion and is incorrect.	
4) Terminal Protein from Adenovirus is linked to a C that serves to prime replication of the linear viral genome. Hence, the last 3'-nucleotide in each end of its linear genome must be:	
A) A nucleotide B) C nucleotide C) G nucleotide D) T nucleotide Correct answer: C) G nucleotide In adenovirus DNA replication, the terminal protein (TP) is covalently linked to a dCMP (a C nucleotide). This C serves as the primer for DNA polymerase. For DNA synthesis to initiate, the priming C nucleotide on the terminal protein must base-pair with the nucleotide at the 3' end of the viral DNA template. Because base pairing follows the rule: • C pairs with G the template strand must end with a G at its 3' terminus so that the priming C can properly pair and replication can begin. Therefore, the last 3' nucleotide at each end of the linear adenovirus genome must be G.	
Why the other answers are wrong A) A nucleotide A pairs with T, not C. A terminal C primer cannot pair with A, so replication cannot start. B) C nucleotide C pairs with G, not with itself. A 3' terminal C cannot serve as the template for a priming C. D) T nucleotide	

T pairs with A. This also cannot base-pair with a priming C nucleotide.	
5) The mitochondrial chromosome is composed as A) One linear chromosome B) Several linear identical chromosomes C) One circular chromosome D) Several identical circular chromosomes Correct answer: D) Several identical circular chromosomes Mitochondria contain their own genome, which is organized as multiple copies of a small circular DNA molecule. Each mitochondrion typically carries many identical copies of this circular chromosome, and each cell contains many mitochondria, further increasing the copy number. This high redundancy ensures reliable mitochondrial function and allows the organelle to maintain sufficient gene expression even when some copies become damaged. Thus, the mitochondrial genome is best described as several identical circular chromosomes, making D the correct answer.	
Why the other answers are wrong A) One linear chromosome Mitochondrial DNA is not linear. Linear DNA is characteristic of nuclear chromosomes in eukaryotes, not mitochondria. B) Several linear identical chromosomes Although mitochondria contain multiple genome copies, they are circular, not linear. Therefore, this option is incorrect because of the wrong DNA shape. C) One circular chromosome While mitochondria do contain a circular chromosome, they do not contain just a single copy. Each mitochondrion carries many copies, and each cell contains many mitochondria. Thus, this answer underestimates the number of mitochondrial genomes.	
6) An IS sequence is flanked by a short direct repeat derived from duplication of A) Transposon DNA B) DNA at the target site C) de novo synthesis (like telomeres) D) DNA from another transposon Correct answer: B) DNA at the target site An IS (insertion sequence) element inserts itself into a new location in the genome by a transposition mechanism that involves staggered cutting of the target DNA. When the IS element is inserted, the cell's DNA repair machinery fills in the single-stranded gaps created by the staggered cut. This filling-in produces a short duplication of the target site DNA, and these duplicated sequences appear as direct repeats flanking the newly inserted IS element.	

Thus, the flanking repeats do not come from the transposon itself—they are copied from the target DNA.	
Why the other answers are wrong A) Transposon DNA The direct repeats do <i>not</i> originate from the transposon sequence. The transposon inserts between target nucleotides, but the repeats are made from the host DNA that is duplicated during the repair process. Thus, this option is incorrect. C) de novo synthesis (like telomeres) The short direct repeats are not synthesized anew in the way telomeric repeats are. They arise specifically from copying the target site DNA, not from a telomerase-like mechanism of building a repeat from scratch. D) DNA from another transposon Each insertion event produces repeats based on the local target sequence, not from other transposons elsewhere in the genome. No horizontal copying from other transposons occurs in this process.	
7) A retrovirus may have evolved from	
A) a retrotransposon B) a dsDNA based virus C) an ssDNA based virus D) a recombinase Correct answer: A) a retrotransposon Retroviruses are widely believed to have evolved from retrotransposons, specifically from elements resembling LTR (long-terminal-repeat) retrotransposons. Retrotransposons and retroviruses share key molecular features: • Both replicate through an RNA intermediate. • Both encode reverse transcriptase and integrase, allowing them to convert RNA into DNA and integrate it into the host genome. • Both use LTRs to control transcription and integration. The main difference is that retroviruses have acquired genes that enable them to form viral particles and leave the cell, most importantly the env gene. This suggests that a retrotransposon gained additional functions allowing extracellular transmission, evolving into a retrovirus. Thus, A is correct.	
Why the other answers are wrong B) a dsDNA based virus Double-stranded DNA viruses do not use reverse transcription, do not rely on RNA intermediates, and lack the machinery characteristic of retroviruses. Their replication strategies are fundamentally different, so they are unlikely to be ancestral to retroviruses.	
C) an ssDNA based virus	

Single-stranded DNA viruses replicate through DNA intermediates, not RNA intermediates, and do not encode reverse transcriptase. Their life cycle does not resemble retroviral replication, making them an implausible evolutionary origin.	
D) a recombinase A recombinase is a protein, not a self-replicating genetic element. Proteins cannot “evolve into” viruses. Although integrase in retroviruses is related to recombinases, a recombinase cannot serve as a precursor of an entire viral genome.	
8) The virion of a retrovirus contains how many RNA strands?	
A) One B) Two C) Three D) Four Correct answer: B) Two A retroviral particle (virion) contains two copies of its single-stranded RNA genome. These two RNA molecules are identical and are packaged together as a dimer, held in close association by base-pairing and viral proteins. This diploid nature of the retroviral genome is a hallmark of retroviruses and provides advantages such as recombination and template switching during reverse transcription, increasing genetic robustness and variability. Thus, a retroviral virion contains two RNA strands, making B the correct answer.	
Why the other answers are wrong A) One Although many RNA viruses package a single RNA genome, retroviruses do not. They are uniquely diploid, always packaging a pair of RNA molecules. Therefore, one RNA strand is insufficient for a retroviral virion.	
C) Three No retrovirus packages three RNA strands. Retrovirus structure and replication rely on having a dimeric genome, not more.	
D) Four Similarly, retroviruses never package four RNA strands. Such a configuration would not fit with the reverse transcription mechanism or known virion structures.	
9) How many template switches (jumps) are required to complete turning the infective RNA strand of a retrovirus into DNA	
A) One B) Two C) Three	

D) Four Correct answer: B) Two During reverse transcription in a retrovirus, the viral reverse transcriptase converts the single-stranded RNA genome into double-stranded DNA. Because the virion contains two identical RNA strands, reverse transcription naturally involves template switching—also known as “jumps”—between these two RNA molecules. To complete the full synthesis of the proviral DNA, two template switches are required: 1. First jump (during minus-strand synthesis): After the primer and initial DNA synthesis at the 5' end of the RNA genome, the newly synthesized short DNA fragment transfers to the 3' end of the RNA genome to continue elongation. 2. Second jump (during plus-strand synthesis): After part of the plus strand is made, the DNA switches templates again, enabling synthesis across both ends of the genome to complete the full-length dsDNA. These two template switches are essential for reconstructing the complete linear DNA version of the viral genome, including both LTRs. Thus, the correct answer is two.	
Why the other answers are wrong A) One A single template switch is not enough to reconstruct both ends of the viral DNA or generate the proper long terminal repeats. Retroviral DNA synthesis universally requires two jumps. C) Three Three jumps are not part of the standard reverse transcription mechanism. Additional switches do not occur and are not necessary for completing viral DNA synthesis. D) Four Four template switches would be incompatible with known retroviral replication mechanisms and are never observed. Reverse transcriptase performs exactly two essential template jumps.	
10) The 2-micron circle in yeast can alter between regular replication mode and double-rolling circle mode due to ...	
A) a stochastic event B) direct repeat recombination induced by the Frt recombinase C) Inverted repeat recombination induced by the Frt recombinase D) non-homologous recombination induced by the Frt recombinase Correct answer: C) inverted repeat recombination induced by the Frt recombinase	

The 2-micron plasmid in yeast contains two FRT sites arranged as inverted repeats, not direct repeats. Because they are inverted relative to one another, recombination between them catalyzed by the Flp recombinase results in inversion of the DNA segment between the FRT sites rather than excision. This inversion event toggles the orientation of a key replication-control region on the plasmid. When the segment is in one orientation, the plasmid replicates using standard bidirectional replication. When the Flp recombinase inverts the region, the plasmid switches into double-rolling-circle replication, which dramatically increases plasmid copy number. Thus, the ability of the 2-micron circle to alternate between replication modes is specifically due to recombination between inverted FRT repeats, making C the correct answer.

Why the other options are wrong

A) a stochastic event

The switch is not random. It is a regulated, enzyme-driven recombination reaction controlled by the Flp recombinase and the sequence arrangement of the FRT sites.

B) direct repeat recombination induced by the Frt recombinase

Recombination between direct repeats would remove or excise the DNA between the repeats as a circle.

This is *not* what happens in the 2-micron plasmid.

The plasmid maintains its structure and only inverts the intervening region, which requires inverted, not direct, repeats.

D) non-homologous recombination induced by the Frt recombinase

Flp recombinase is a site-specific recombinase that requires identical FRT sequences.

It does not perform non-homologous recombination or NHEJ; therefore, this option is incorrect.

Task set B

Task set B contains 2 essay questions (4 in the final exam). The answer should be written using the space allocated below each question.

- 1) In the context of a retrovirus; explain why it is difficult to replicate a linear DNA molecule.

DNA polymerase cannot fully copy the ends of a linear DNA molecule because of a fundamental limitation in how DNA synthesis works. DNA polymerases can only add nucleotides to an existing 3'-OH group; they cannot start synthesis on their own. During replication, the leading strand can be copied continuously, but the lagging strand must be synthesized in short Okazaki fragments, each beginning with an RNA primer. At the very end of a linear chromosome, the final RNA primer sits at the extreme 5' end of the lagging strand. When this primer is removed, there is no upstream 3'-OH group available for DNA polymerase to extend from. As a result, this last portion of DNA cannot be replaced, leaving the newly synthesized DNA strand incomplete and shorter than the template. This inherent inability to fill in the terminal gap is known as the end-replication problem. For a retrovirus, this means that its linear DNA would lose essential terminal sequences if it attempted to replicate independently. By integrating into the host genome, the virus avoids this problem entirely, because the host cell's chromosome-maintenance systems protect and replicate linear DNA ends using telomeres and telomerase.

Esperanza's answer:

The fact that DNA polymerases (including Reverse transcriptase) only catalyze polymerization in the 5' to 3' direction combined with the fact that they need a primer to initiate polymerization means that they cannot replicate the very 3'-ends of linear DNA molecules. Hence, special tricks are required to replicate linear chromosomes, linear plasmids, linear transposon RN, intermediates and linear viral genomes.

- 2) It is difficult to combat retroviruses with drugs as they mutate fast to develop resistant variants. An important drug against the HIV virus is an inhibitor of the HIV protease and the virus does not easily develop resistance towards this drug. Speculate why the HIV virus cannot easily develop resistance against drugs that are acting as competitive inhibitors, that targets the HIV protease.

HIV normally develops drug resistance quickly because most antiviral drugs target proteins in ways that allow the virus to accumulate mutations without destroying the protein's essential function.

However, competitive inhibitors of the HIV protease are an important exception, and the virus

cannot easily develop resistance against them. The reason lies in the extremely tight structural constraints on the protease's active site.

HIV protease is essential for viral maturation, and its active site must recognize and cleave a very specific set of peptide sequences in the viral polyprotein. Because this specificity is crucial, the shape and chemical properties of the active site cannot change much without destroying protease function altogether. A competitive inhibitor works by resembling the natural substrate and binding directly in that active site.

For the virus to become resistant, it would need mutations that prevent the inhibitor from binding but still allow the natural substrate to fit perfectly.

This is extremely difficult because even small mutations in the active site tend to:

- **disrupt substrate binding,**
- **reduce catalytic activity,**
- **impair cleavage of viral proteins, and**
- **produce noninfectious virions.**

In other words, the active site of HIV protease is so finely tuned that it cannot mutate to exclude the inhibitor without fatally compromising its own function.

Thus, the virus has little evolutionary room to escape a competitive inhibitor targeting this highly constrained region. This structural rigidity explains why HIV develops resistance much more slowly to protease inhibitors than to drugs targeting other viral proteins, such as reverse transcriptase.

Esperanza's answer:

The HIV protease is essential for the development of all proteins required to complete the lifecycle of HIV. Hence inhibition of the protease abolishes viral infection. It is probably easy

for the virus to mutate the protease to form variants that do no longer bind the competitive inhibitor. Typically, and most frequently, the virus will mutate the active site, hence, changing the binding specificity of the protease. However, the consequence of the changed specificity of the protease would also mean that it no longer has the ability to process the viral proteins correctly. Hence, to combat the effect of the drug, the virus needs to produce several successful mutations simultaneously. It needs to produce at least one mutation that changes the binding specificity of the protease. At the same time the virus also need to produce mutations that alter the cleavage-sites of the protease to sequences that match the new specificity of the mutant protease – and this will happen very rarely.

Task set C

Main Question 1

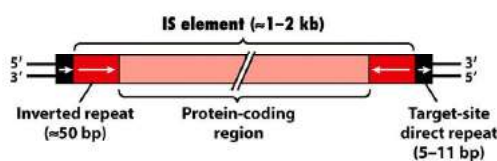
Question 1: Class II composite transposons are flanked by IS or IS-like elements. Why does transposition not depend on the orientation of the two IS1 elements (they can be organized as direct or inverse repeats)

Composite transposons do not depend on whether their two flanking IS elements are arranged as direct or inverted repeats because transposition uses only the outward-facing terminal ends of the IS elements, which form a functional substrate regardless of internal orientation. The transposase does not require any specific relationship between the inner ends, so both arrangements allow efficient mobilization.

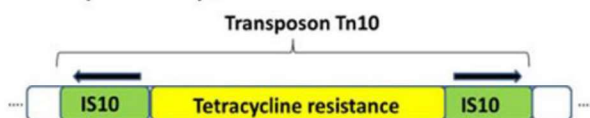
Esperanza's answer:

The IS sequences are themselves flanked by inverted repeats. These repeats define the excision sites. Hence, the orientation does not matter.

Simple Transposon

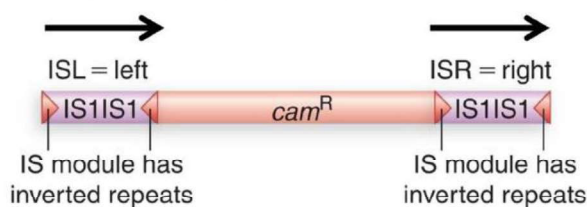


Composite Transposon



Transposon Tn9

Composite Transposon



Question 2. It is good for the selfish transposon to be propagated by the host. But they may also help the host. What would benefit a bacteria infected with Tn10?

A bacterium infected with Tn10 gains a clear selective advantage because Tn10 carries a tetracycline-resistance gene.

Here is the explanation in coherent form:

Answer

Although Tn10 is a *selfish* genetic element whose primary “goal” is to propagate itself, it can simultaneously benefit the bacterial host. Tn10 encodes a tetracycline-resistance gene, allowing the bacterium to survive and grow in environments that contain tetracycline. This gives the infected bacterium a major survival advantage over bacteria that lack Tn10.

Thus, while the transposon spreads because it moves between DNA molecules, the host cell also gains a strong selective advantage in antibiotic-containing environments, enabling the bacterium to thrive where others would be killed.

Summary

➔ **A bacterium carrying Tn10 benefits by becoming resistant to tetracycline.**

This improves the bacterium’s survival and ensures propagation of both the host and the transposon in antibiotic environments.

Esperanza’s answer:

They would be resistant to tetracyclin

Question 3. Tn9 and Tn10 species can be lost from the genome, but not with the same rate. In fact Tn9 is lost with a much higher rate than Tn10, which is in fact quite stable. Propose a model why this is the case.

Tn9 and Tn10 differ in how stably they are maintained in the bacterial genome, and this difference can be explained by the behavior of their flanking IS elements. Both transposons are composite elements, meaning that the antibiotic-resistance gene in the center is flanked by two IS modules. These IS elements can recombine with each other, and if they do, the entire transposon between them can be precisely excised from the chromosome. The likelihood of such excision depends on how readily the two IS elements recombine.

Tn9 is flanked by IS1 elements, which recombine efficiently. Because IS1 pairs easily and undergoes frequent recombination, the central region carrying the resistance gene is often deleted from the genome. As a result, Tn9 is lost at a relatively high rate.

In contrast, Tn10 is flanked by IS10 elements, which recombine much less efficiently. IS10 is tightly regulated and forms a more stable composite structure that rarely undergoes recombination events capable of excising the transposon. Consequently, Tn10 remains stably integrated in the genome.

Thus, the much higher loss rate of Tn9 compared to Tn10 can be explained by the tendency of IS1 to recombine readily, whereas IS10 does so only rarely.

Esperanza's answer:

The IS sequences in Tn9 are organized as direct repeats. Spontaneous direct repeat recombination between these two repeats will eliminate the *camR* gene and either the ISL or the ISR repeat. With IS10 the IS repeats are inverted repeats. In this case recombination between IS sequences just leads to inversion. Several types of DR repeat recombination event could loop out Tn10, but these repeats are much smaller.

Question 4. Explain how Tn10 can be lost from the genome in a very rare recombination event.

Tn10 is lost when its two flanking IS10 sequences undergo a rare homologous recombination event. This recombination excises the entire transposon as a circular DNA molecule and leaves behind a single IS10 copy in the chromosome. Although the event is very uncommon, it is the principal mechanism by which Tn10 can disappear from the genome.

So, the rare loss of Tn10 from the genome has a specific name. It is called precise excision, and it occurs through homologous recombination between the two flanking IS10 elements, which act as direct repeats. When these nearly identical IS10 sequences recombine, the entire Tn10 transposon is removed in one clean event, restoring the chromosome to exactly the state it was in before Tn10 was inserted. Only a single IS10 element remains in the chromosome. Because the excision is accurate and leaves no mutations behind, it is referred to as a *precise* excision.

Esperanza's answer:

Recombination between the short direct repeats of host DNA will lead to loss of the transposon.

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Assignment 10. DNA repair and recombination

Task set A

Task set A contains 10 multiple choice questions (20 in final exam). The correct answer should be indicated by a letter in the cell next to the possible answers.

	correct answer
1) A Repair of DNA DSBs requires a template. For a single DNA DSB approximately in the middle of yeast chromosome I; which of the following templates can be applied for this purpose ?	
A) A sequence from Chromosome I B) A sequence from Chromosome II C) Telomeres from Chromosome I D) Telomeres from Chromosome II Correct Answer: A) A sequence from Chromosome I Repair of DNA double-strand breaks (DSBs) by homologous recombination requires a homologous DNA template. In budding yeast (as in other eukaryotes), the repair machinery searches for a DNA sequence that is highly homologous and physically available to guide accurate repair. If the break occurs in the middle of chromosome I, the only suitable template is a homologous region from the same chromosome—that is, chromosome I itself (typically the sister chromatid if the cell is in S or G2 phase, or the homolog in diploids). A sequence from the <i>same</i> chromosome provides the needed sequence identity and correct genomic context for high-fidelity repair.	
Why the other options are wrong B) A sequence from Chromosome II Chromosome II, like all nonhomologous chromosomes, carries completely unrelated DNA sequence in the region corresponding to the break. Because homologous recombination requires near-perfect sequence homology, a sequence from chromosome II cannot serve as a repair template. Using a nonhomologous chromosome would instead lead to misrepair, translocations, or activation of nonhomologous end joining (NHEJ), not accurate DSB repair. C) Telomeres from Chromosome I Although these come from the <i>same chromosome</i> , telomeres contain telomeric repeat sequences, which are: • far from the break site (the break is in the middle of chromosome I, telomeres are at the ends), • nonhomologous to internal chromosomal sequences, • highly repetitive and unsuitable as a template for repair of an internal DSB. Thus, telomeres cannot provide the necessary homologous sequence for templated repair in the chromosome's interior. D) Telomeres from Chromosome II	

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This option combines the problems of B and C: Not only is chromosome II nonhomologous to chromosome I, but its telomeres also contain repetitive telomeric sequence, which bears no resemblance to the internal sequence where the break occurred. This makes them entirely useless as recombination templates.	
2) If many DNA double strand ends in a cell need to be joined; this can be safely achieved by homologous recombination because ...	
A) DNA ends themselves contain all necessary information to join correctly B) The right partner ends can be identified by matching their sequences to sequences present on intact sister chromatids or homologous chromosomes C) it does not matter, which ends are joined D) Ligation is an error-free reaction Correct Answer: B) The right partner ends can be identified by matching their sequences to sequences present on intact sister chromatids or homologous chromosomes Homologous recombination is a <i>template-guided</i> repair mechanism. When many double-strand breaks (DSBs) occur in a cell, the danger is that incorrect DNA ends could be joined together, generating gross chromosomal rearrangements such as translocations or inversions. Homologous recombination avoids this risk because each broken DNA end searches for and pairs with a highly homologous sequence located on an intact sister chromatid or homologous chromosome. This process of sequence matching—mediated by proteins like Rad51—ensures that only the correct ends engage in repair. Thus, even when many ends are present, homologous recombination prevents accidental misjoining by using homology as the identifier of correct partner ends.	
Why the other answers are wrong A) DNA ends themselves contain all necessary information to join correctly This is incorrect because DNA ends, taken alone, do not carry positional information that would allow them to “know” which other end belongs to the same original break. Without a homologous template, the cell cannot distinguish ends from different breaks. If DNA ends were simply ligated based on their own features, the outcome would be random and highly mutagenic. Homologous recombination works <i>not</i> because the ends direct the repair, but because the template directs the repair.	
C) It does not matter, which ends are joined This is fundamentally wrong. Mismatched end-joining would create: • chromosomal translocations • deletions	

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• dicentric or acentric fragments • loss of genetic information These are catastrophic for genome stability. The entire purpose of homologous recombination is to avoid such misjoining by enforcing sequence homology as a guide. Therefore, it matters enormously which ends are joined.	
D) Ligation is an error-free reaction While ligation as a biochemical step is indeed precise, this does not address the core problem: ligation must connect the correct ends. Even perfect ligation of the <i>wrong</i> ends would produce severe genomic damage. Homologous recombination's accuracy comes not from the ligation reaction itself, but from the homology-directed alignment that precedes it.	
3) Which type of enzyme is central in Synthesis-Dependent Strand-Annealing	
A) DNA polymerase B) Proteinase K C) Telomerase D) Primase Correct Answer: A) DNA polymerase Synthesis-Dependent Strand Annealing (SDSA) is a major homologous recombination-based repair pathway used to fix DNA double-strand breaks without forming crossovers. After one of the resected single-stranded DNA ends invades a homologous template, the key step that extends this invading strand is new DNA synthesis. This synthesis is performed by DNA polymerases, which copy the sequence from the intact homologous duplex. The newly extended strand then dissociates and anneals to the complementary region on the other side of the break, completing the repair. Because DNA synthesis is the central, defining step in SDSA, DNA polymerase is the essential enzyme for this pathway.	
Why the other options are wrong B) Proteinase K Proteinase K is a broad-spectrum protease used experimentally to degrade proteins during sample preparation. It has no physiological role in DNA repair pathways and is not involved in homologous recombination or SDSA. Therefore, it cannot be the central enzyme in SDSA.	
C) Telomerase Telomerase extends telomeres by adding repetitive sequences to chromosome ends. While it is a specialized reverse transcriptase, its activity is restricted to telomere	

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maintenance, not general double-strand break repair. It does not participate in strand invasion, DNA synthesis, or annealing steps required for SDSA.	
D) Primase Primase synthesizes short RNA primers needed to initiate DNA replication. However, SDSA does not require de novo priming of RNA primers at the invasion site. Instead, DNA polymerase can extend the invading 3' end directly using the intact homologous duplex as a template. Primase plays no essential role in this repair mechanism.	
4) In the absence of telomeres, DNA DSB repair may accidentally ...	
A) Break down the centromeres B) Lead to chromosome duplication C) Lead to circularization of chromosomes D) Activate cell division Correct Answer: C) Lead to circularization of chromosomes Telomeres protect chromosome ends by marking them as natural, <i>legitimate</i> termini rather than double-strand breaks (DSBs). If telomeres are absent—because they erode, are experimentally removed, or telomerase is inactive—the chromosome ends are no longer recognized as protected ends. Instead, the cell interprets these exposed ends as DNA double-strand breaks. The DSB repair machinery (especially nonhomologous end joining, NHEJ) may then ligate the two unprotected ends of the same chromosome together, mistakenly treating them as broken DNA ends belonging to a single DSB. This leads to end-to-end joining, producing: • circular chromosomes, or • dicentric chromosomes when two different chromosome ends are joined. Thus, accidental circularization is a characteristic outcome of DSB repair acting on chromosome ends in the absence of telomeres.	
Why the other options are wrong A) Break down the centromeres Centromeres are internal chromosomal structures and are not directly affected by the absence of telomeres. Loss of telomeres does not cause centromeres to be degraded or broken down; instead, the instability occurs specifically at chromosome ends, not at centromeric DNA.	
B) Lead to chromosome duplication Telomere loss does not trigger chromosome duplication. Chromosome duplication is controlled by the DNA replication machinery and cell cycle checkpoints. While dysfunctional telomeres can cause fusion–breakage cycles that eventually lead	

to complex rearrangements, they do not directly cause whole-chromosome duplication as an immediate consequence of DSB repair.	
D) Activate cell division Telomere dysfunction does not activate division; typically it produces the opposite effect. Exposed chromosome ends activate a persistent DNA damage response, which usually leads to: • cell cycle arrest, • senescence, or • apoptosis. Thus, the absence of telomeres impedes division rather than promoting it.	
5) Meiosis results in formation of	
A) many identical haploid cells B) many different haploid cells C) many identical diploid cells D) many different diploid cells Correct Answer: B) many different haploid cells Meiosis is a specialized cell division process that reduces the chromosome number from diploid (2n) to haploid (n) and generates genetic diversity. Two mechanisms make the resulting cells genetically different from one another: 1. Independent assortment of homologous chromosomes mixes maternal and paternal chromosomes in new combinations. 2. Homologous recombination (crossing over) exchanges DNA segments between homologs, creating new allele combinations within chromosomes. Because each meiotic division produces gametes with a unique combination of alleles, meiosis results in many genetically different haploid cells.	
Why the other options are wrong A) many identical haploid cells This describes the outcome of mitosis, not meiosis. Mitosis generates daughter cells that are genetically identical to each other and to the parent cell. Meiosis, by design, introduces genetic diversity; therefore, its products are not identical.	
C) many identical diploid cells This is incorrect for two reasons: • Meiosis does not produce diploid cells; it reduces chromosome number to haploid.	

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Producing identical diploid cells again describes mitotic division, not meiotic division.	
D) many different diploid cells Although meiosis does produce genetically different cells, they are haploid, not diploid. The reduction division (meiosis I) ensures that chromosome number is halved, which is essential for sexual reproduction, so diploid products would not be correct. Difference between diploid and haploid: Diploid cells (2n) Diploid cells contain two complete sets of chromosomes—one inherited from the mother and one from the father. Thus, each chromosome is present in pairs called <i>homologous chromosomes</i>. Characteristics: - They have two copies of every gene, which may be identical (homozygous) or different (heterozygous). - Diploidy allows for dominance relationships, genetic buffering, and recombination between homologs. - Most somatic (body) cells in animals and plants are diploid. - In humans, diploid cells contain 46 chromosomes (23 pairs). Diploidy preserves genetic stability during mitotic divisions and provides the starting material for meiosis.	
Haploid cells (n) Haploid cells contain only one complete set of chromosomes. They therefore have one copy of each gene, with no homologous chromosome present. Characteristics: - They carry half the genetic material of a diploid cell. - Haploidy is essential for sexual reproduction, because fusion of two haploid gametes (egg and sperm) restores diploidy in the zygote. - Meiosis produces haploid cells through reduction of chromosome number. - In humans, haploid cells contain 23 chromosomes. Haploid cells are crucial for generating genetic diversity and maintaining the species' chromosome number across generations.	
6) As a single diploid cell proceeds through the two consecutive meiotic divisions I and II, the result is formation of	
A) one haploid cell B) two haploid cells C) four haploid cells	

D) eight haploid cells Correct Answer: C) four haploid cells A single diploid cell ($2n$) entering meiosis undergoes two sequential divisions: meiosis I and meiosis II. 1. Meiosis I is the <i>reductional division</i>: Homologous chromosomes separate, reducing the chromosome number from diploid ($2n$) to haploid (n). → This produces two haploid daughter cells. 2. Meiosis II is the <i>equational division</i>: Sister chromatids separate, much like in mitosis but occurring in haploid cells. → Each of the two haploid products from meiosis I divides again, yielding a total of four haploid cells. Thus, the defining outcome of meiotic division is the production of four genetically distinct haploid cells from one diploid precursor.	
Why the other options are wrong A) one haploid cell This is incorrect because meiosis consists of <i>two</i> rounds of division, not one. A single haploid product would imply the process stopped after meiosis I, which does not occur in normal meiosis.	
B) two haploid cells This number corresponds only to the products of meiosis I. But meiosis continues into meiosis II, where each of these two haploid cells divides again. Therefore, the final outcome cannot be only two haploid cells.	
D) eight haploid cells Eight haploid cells would require three rounds of division or meiotic replication beyond what actually happens. Meiosis always consists of exactly <i>two</i> divisions following a single round of DNA replication. Thus, eight products cannot result from a single meiotic event.	
7) In many cases, a DNA double strand break may be sealed correctly by non-homologous end-joining because ...	
A) right after the break, ends are captured and held together by the KU Hetero dimer and the MRX complex B) DNA ligase is error free C) it uses homologous recombination D) sequences at the ends always contain long stretches of homologous sequences Correct Answer: A) right after the break, ends are captured and held together by the KU heterodimer and the MRX complex Non-homologous end joining (NHEJ) is a DNA repair pathway that does not rely on sequence homology. Its accuracy—when it succeeds—is largely due to the fact that the correct DNA ends are physically preserved and protected immediately after the break occurs. Two protein assemblies play a key role:	

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• KU70/80 heterodimer (KU complex) binds the broken DNA ends with high affinity, shielding them from degradation and preventing them from drifting away. • The MRX (MRN in mammals) complex helps tether the ends and organizes the repair machinery. Because the proper ends are physically held together from the moment the damage occurs, NHEJ can often re-ligate them correctly simply by keeping the right partners in place, even though it does not check for sequence similarity. This structural preservation—not sequence information—is what allows NHEJ to be accurate enough in many cases.	
Why the other answers are wrong B) DNA ligase is error free The act of ligation itself is chemically precise, but this does not ensure correct repair. If the wrong ends are presented to DNA ligase, it will just as readily join them, causing translocations or deletions. The fidelity of NHEJ comes from correct end pairing (mediated by KU and MRX), <i>not</i> from any special error-free property of ligase.	
C) it uses homologous recombination This is incorrect because NHEJ is defined by the absence of homology dependence. It is fundamentally distinct from homologous recombination (HR), which requires a homologous template. NHEJ is template-independent and therefore does not use HR at all.	
D) sequences at the ends always contain long stretches of homologous sequences If this were true, NHEJ would be unnecessary. In reality, most breaks do not expose perfectly homologous ends, and NHEJ operates precisely because it does not require homology. When long stretches of homology <i>are</i> present, the cell preferentially uses homologous recombination, not NHEJ.	
8) How many DNA strands can act as template for HR based DNA Repair in A diploid cell after DNA replication	
A) One B) Two C) Three D) Four Correct Answer: C) Three After DNA replication in a diploid cell, there are four chromatids that contain homologous sequence for any given locus: 1. The two sister chromatids of the maternal homolog 2. The two sister chromatids of the paternal homolog → Total = 4 chromatids	

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When one of these chromatids suffers a double-strand break, the two strands at the break site can <i>no longer</i> serve as a template — they are damaged and lack continuity of sequence. Therefore, they cannot guide homologous recombination–based repair. That leaves three intact chromatids, each of which carries a correct, continuous copy of the DNA sequence at that locus: • the unbroken sister chromatid (1 template) • the two chromatids of the homologous chromosome (2 templates) → Total intact templates = 3 Homologous recombination can use any of these three intact DNA molecules as a sequence template, although in practice the intact sister chromatid is overwhelmingly preferred due to its proximity and perfect identity.	
Why the other answers are wrong A) One This would only be true in a haploid cell after replication (one intact sister chromatid). In a diploid cell, more homologous copies remain available.	
B) Two This would be correct only if the homologous chromosome could not serve as a template. In reality, both chromatids of the homolog contain usable homologous sequences.	
D) Four This would be true <i>only if none of the chromatids were broken</i>. But in the context of DSB repair, the broken chromatid cannot serve as its own template, so the maximum number of usable templates is three.	
9) Rad52 directly stimulates	
A) Binding of RPA to Double stranded DNA B) DNA annealing C) DNA helicase activity D) Holliday Junction resolution Correct Answer: B) DNA annealing Rad52 is a central recombination mediator protein in yeast and plays a crucial role in homologous recombination pathways, especially those involving single-strand annealing and post-synaptic steps of repair. Its hallmark biochemical activity is its ability to directly promote the annealing of complementary single-stranded DNA molecules.	

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Rad52 forms ring-shaped oligomers that bind ssDNA and bring complementary strands into proximity, thereby stimulating spontaneous DNA pairing even when RPA is still present. This annealing activity is essential for processes such as: • Single-strand annealing (SSA) • Second-end capture during homologous recombination • Rad51-independent recombination pathways Because DNA annealing is the reaction Rad52 directly catalyzes, B is the correct answer.	
Why the other answers are wrong A) Binding of RPA to double-stranded DNA Rad52 does not stimulate RPA binding to dsDNA. RPA (Replication Protein A) primarily binds single-stranded DNA, not double-stranded DNA, and Rad52's relationship with RPA is instead competitive and regulatory. Rad52 helps displace RPA from ssDNA to allow Rad51 filament formation—not binding RPA to dsDNA. Thus, this option is incorrect both mechanistically and conceptually.	
C) DNA helicase activity Rad52 is not a helicase and has no ATP-dependent unwinding activity. Helicases such as Sgs1 (BLM in mammals) or Rad3 family helicases perform unwinding steps in recombination and DNA repair, but Rad52 does not. Therefore, Rad52 does not stimulate helicase activity, nor does it possess any itself.	
D) Holliday Junction resolution Rad52 plays no direct role in resolving Holliday junctions. This process is carried out by specialized structure-specific endonucleases such as: • Mus81–Mms4 • Yen1 • Slx1–Slx4 Rad52 acts much earlier in recombination (strand annealing and mediator functions), so it is not involved in junction resolution. Thus, option D is incorrect.	
10) A Holliday Junction is able to ...	
A) Initiate strand invasion DNA strand together B) Branch migrate C) Initiate DNA polymerization D) Prevent DNA crossing over Correct Answer: B) Branch migrate A Holliday junction is a four-stranded DNA intermediate formed during homologous recombination.	

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Its defining structural property is that the junction can move along the DNA duplexes without breaking base-pairing. This movement is called branch migration. During branch migration, the crossover point slides along the DNA, effectively enlarging or shifting the region of heteroduplex DNA. This property is intrinsic to Holliday junctions and is essential for recombination outcomes, influencing gene conversion and crossover patterning. Thus, branch migration is a fundamental and characteristic activity of Holliday junctions.	
Why the other answers are wrong A) Initiate strand invasion A Holliday junction does <i>not</i> initiate strand invasion. Strand invasion occurs before the Holliday junction is formed and is catalyzed by proteins such as Rad51 in eukaryotes or RecA in bacteria. Once strand invasion and DNA synthesis occur, <i>then</i> a Holliday junction may form. Thus, the junction is a product, not an initiator, of strand invasion.	
C) Initiate DNA polymerization Holliday junctions do not start DNA synthesis. DNA polymerization during homologous recombination begins right after strand invasion, using the invading 3' end as a primer and the intact homologous chromatid as a template. By the time a Holliday junction appears, the polymerization step has already taken place. The junction itself has <i>no enzymatic ability</i> to initiate DNA synthesis.	
D) Prevent DNA crossing over This is the opposite of reality. Holliday junctions are the structural intermediates that allow crossing over to occur. Depending on how the junctions are resolved (and whether they undergo dissolution), they may generate either: • crossover products (exchange of flanking regions) • non-crossover products But they certainly do not <i>prevent</i> crossing over; they are essential for enabling it.	

*A diploid cell has a pair of homologous chromosomes. After DNA replication, a chromosome consists of two sister chromatids. Hence if the cell has a single DNA DSB it can use the intact sister chromatid as template for repair. In addition, it can use the two intact sister chromatids of the homologous chromosome

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Task set B

Task set B contains 2 essay questions (4 in the final exam). The answer should be written using the space allocated below each question.

1) Why do we, in addition to non-homologous end-joining (NHEJ), need a pathway based on homologous recombination (HR) for the repair of DNA double-strand breaks (DNA DSBs)?

Cells require two distinct pathways—non-homologous end-joining (NHEJ) and homologous recombination (HR)—to repair DNA double-strand breaks (DSBs), but these pathways do not act sequentially. NHEJ does not repair first and then hand the break to HR “if needed.” Instead, the two pathways compete for the break from the moment it occurs, and the cell must choose one of them. NHEJ acts extremely quickly. The KU heterodimer binds the broken DNA ends within seconds, protecting them and allowing fast repair. This speed is crucial, particularly in G1 phase, where no sister chromatid is available. However, because NHEJ simply rejoins the ends without using a template, it is inherently error-prone and can introduce small insertions or deletions. If multiple breaks occur, NHEJ may even join the wrong ends together, leading to harmful chromosomal rearrangements such as translocations or fusions.

For this reason, cells also need HR—a slower but **high-fidelity(= high accuracy, low error rate, faithful restoration of the original DNA sequence.)**, template-based repair pathway. HR uses an intact homologous sequence, typically the sister chromatid available during S and G2 phases, to restore the DNA precisely. This allows accurate repair even when DNA ends are damaged or missing nucleotides. HR is also essential for repairing one-ended breaks that arise when replication forks collapse, a type of damage that NHEJ cannot resolve correctly.

The choice between NHEJ and HR is controlled early after the break occurs. If the cell initiates DNA end resection, producing 3' single-stranded DNA necessary for Rad51 loading, the break is committed to HR and NHEJ is no longer possible. If the ends are kept unresected, NHEJ proceeds. Because resection and NHEJ require mutually incompatible end structures, the two pathways cannot operate one after the other.

Thus, NHEJ and HR are not sequential steps but alternative solutions that address different cellular needs: NHEJ provides rapid emergency repair, while HR provides accurate, template-based restoration of genetic information. Together, they ensure both immediate survival and long-term genomic stability.

Esperanzas answer:

Double strand breaks are (unless the ends contain other types of damage) actually sealed correctly and quite efficiently by NHEJ because the ends are quickly captured by the KU hetero-dimer and the MRX complex right after the break takes place. The concentration of especially the KU complex is quite high in the nucleus. However, if several breaks occur in the cell and the KU/MRX complexes do not capture the ends right after strand cleavage, it is impossible for the cell to figure out, which ends should be joined by NHEJ. If the wrong choices are made the consequence could be e.g. circular chromosomes, translocations, or di-centric chromosomes. In this case (and for these reasons) HR is used for break sealing. In this pathway the sequences of the ends are tested against

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an intact identical (on the sister chromatid) or homologous (on a homologous chromosome) sequence before they are joined. In this way error-free repair is achieved

2) Mitotically growing yeast cells pass through four phases in the cell cycle. Firstly, they enter a gap phase, G1, where they collect sufficient nutrition to allow them to replicate their chromosomes. Replication takes place in the following phase, which is called S-phase. After S-phase, the cells enter a new gap phase, G2, where they gather enough nutrition to divide. Hence, G2 is followed by the M phase where the two sister chromatids are distributed into the two new cells by the process of mitosis. After mitosis, each of the two new cells will then be in G1 to complete the cycle. In haploid cells, repair of DNA DSBs by homologous recombination is not possible in one of the four phases of the cell cycle. Which cell phase?; and why?

In haploid yeast cells, homologous recombination (HR) cannot occur during the **G1 phase** of the cell cycle. HR requires an intact homologous DNA template to accurately repair a double-strand break, and in haploid cells the only available template is the sister chromatid. However, a sister chromatid does not exist in G1, because the chromosomes have not yet been replicated. The cell contains just one copy of each chromosome, with no homologous partner available. As a result, a DNA double-strand break that occurs in G1 cannot be repaired by HR and must instead be handled by non-homologous end-joining (NHEJ). Once the cell enters S-phase and replication produces a sister chromatid, the necessary template becomes available, and HR can proceed in the later phases of the cycle.

Ezperansas Answer:

Repair of a DNA DSB by HR is not possible in G1 cells. There is no template to support repair. A template will first be available in S phase where the DNA is copied and a chromosome for that reason is composed by two identical sister chromatids (of which one can serve as template for repair of the DNA DSB)

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Task set C

Main Question 1

Imagine that three genes in baker's yeast, *GEN1*, *GEN2*, and *GEN3*, are located on the same chromosome. A fourth gene, *GEN4* was located on a different chromosome. Diploids were then made by mating two haploid parental strains of which one contained wild-type alleles of all four genes whereas the other contained four mutant alleles, which all cause a change in phenotype. By analysing the meiotic offspring of the diploid strain, it turned out that the chance of a crossing over between the genes *GEN1* and *GEN2* was essentially 0 %. In contrast the chance of a crossing over between *GEN1* and *GEN3* was 50%.

By doing this experiment, Mendel's second law stating that a heterozygous diploid strains produces gametes where the two genes sorts independently.

Question 1. When the haploid spore-colonies products were analysed for phenotype describing *GEN1*, *gen1*, *GEN4* and *gen4*. Since all genotypes produce an easy to detect phenotype you can quickly determine the genotype of each spore colony by visual inspection. When you describe the colony genotype produced by these alleles, then what combinations did you get? and in what proportions?

Each haploid spore colony carries one allele of *GEN1* and one allele of *GEN4*.

In the diploid parent, these genes are in the combination:

- Chromosome A: **GEN1 GEN4**
- Chromosome A': **gen1 gen4**

This means the diploid is heterozygous at both loci:

GEN1 / *gen1* and *GEN4* / *gen4*.

Because *GEN1* and *GEN4* are on different chromosomes, they assort independently at meiosis (Mendel's second law). During meiosis I, the orientation of the two homologous chromosome pairs is random, so the four possible combinations of alleles in the gametes occur with equal probability.

So, among the haploid spores you will see four phenotypic/genotypic classes:

1. **GEN1 GEN4**
2. **GEN1 gen4**
3. **gen1 GEN4**
4. **gen1 gen4**

Each of these combinations appears at a frequency of about:

- **25% GEN1 GEN4**
- **25% GEN1 gen4**
- **25% gen1 GEN4**
- **25% gen1 gen4**

In other words, the four possible allele combinations for *GEN1* and *GEN4* occur in a 1 : 1 : 1 : 1 ratio among the spore colonies.

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Question 2. Same question as in question 1, but now you consider segregation of *GEN1* and *GEN2*.

Because *GEN1* and *GEN2* are on the same chromosome and the experiment showed that the chance of crossing over between them is essentially 0%, these two genes are completely linked. That means they are always inherited together as a fixed pair.

In the diploid, the arrangement of alleles on the homologous chromosomes is:

- **One chromosome: *GEN1 GEN2***
- **The other chromosome: *gen1 gen2***

So the diploid is *GEN1 GEN2* / *gen1 gen2*, with the alleles in coupling.

Since there is no recombination between *GEN1* and *GEN2*, meiosis can only produce gametes that carry exactly one of the two *parental* combinations:

- ***GEN1 GEN2***
- ***gen1 gen2***

You will never see *GEN1* together with *gen2*, nor *gen1* together with *GEN2* in the spores, because that would require a crossover between *GEN1* and *GEN2*, and the experiment tells us that this does not happen.

Therefore, when you analyse the haploid spore colonies for the *GEN1/GEN2* phenotypes, you will get only two classes:

- **50% of the spores: *GEN1 GEN2***
- **50% of the spores: *gen1 gen2***

So the observed segregation pattern is a 1 : 1 ratio of the two parental combinations, with no recombinant types.

Question 3. Same question as in question 1, but now you consider segregation of *GEN1* and *GEN3*.

Because *GEN1* and *GEN3* lie on the same chromosome, but the experiment showed that the chance of crossing over between them is 50%, they behave as if they were on different chromosomes. A recombination frequency of 50% is the maximum possible value and means that meiosis produces recombinant and parental combinations in equal proportions—the genes assort independently.

Arrangement of alleles in the diploid

The parental diploid has homologous chromosomes carrying:

- **One chromosome: *GEN1 GEN3***
- **The other chromosome: *gen1 gen3***

Thus, the diploid is in coupling phase:

GEN1–GEN3 / *gen1–gen3*

A 50% recombination frequency means that half of the gametes are parental types and half are recombinant types.

The four possible genotypes among haploid spores

Parental (non-recombinant) combinations

Appear with 50% total, that is 25% each:

- ***GEN1 GEN3***

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- **gen1 gen3**

Recombinant combinations

Also appear with 50% total, that is 25% each:

- **GEN1 gen3**
 - **gen1 GEN3**
-

Final proportions

So when you score spore colonies for GEN1 and GEN3 phenotypes, you obtain:

- **25% GEN1 GEN3 (parental)**
- **25% gen1 gen3 (parental)**
- **25% GEN1 gen3 (recombinant)**
- **25% gen1 GEN3 (recombinant)**

A perfect 1 : 1 : 1 : 1 ratio.

Question 4. The experiments above show that Mendel's second law sometime do not apply to describe the outcome of a cross. In genetic language, it is called non-Mendelian inheritance. Based on the experiments, define a rule stating when Mendel's law does not work.

Mendel's second law fails when two genes are located so close together on the same chromosome that crossing over between them is rare or absent.

Explanation

Mendel's second law (independent assortment) states that alleles of different genes segregate independently during gamete formation. This is true when:

- the genes are on different chromosomes, or
- they are far apart on the same chromosome with frequent recombination.

However, the experiments show two exceptions:

1. **GEN1 and GEN2 showed 0% recombination**

→ They always cosegregate.

→ Mendel's second law does *not* describe their behavior.

2. **GEN1 and GEN3 showed 50% recombination**

→ They assort as if on different chromosomes and *do* follow the law.

This demonstrates that physical linkage can override Mendel's expectations.

Final Rule:

Mendel's second law does not apply to genes that are physically linked—that is, genes located so close together on the same chromosome that crossing over between them occurs at a frequency significantly less than 50%. Such genes do not assort independently but instead are inherited together more often than predicted by Mendelian ratios.

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Question 5. If you have a diploid strain with the genotype *GEN5/gen5* and *GEN6/gen6*, which produces offspring with all possible strain combination in ratios 25%:25%:25%:25%; can you conclude that the two genes are positioned on two different chromosomes?

No, you cannot conclude that *GEN5* and *GEN6* are on different chromosomes. A 25%:25%:25%:25% ratio only shows that the genes assort independently. This can happen either because they are on different chromosomes or because they are far apart on the same chromosome and recombine at 50%. Therefore, the observed 25%:25%:25%:25% ratio alone cannot determine whether the genes are on separate chromosomes or simply far apart on the same chromosome; additional mapping or physical evidence would be needed to distinguish between these possibilities.

Main Question 2

In the mechanism for HR repair of DNA DSBs, the ends of the break are initially processed by exonucleases to produce long 3'-OH single-stranded tails. Next these tails are covered by the single strand binding protein complex RPA. Such ends do not engage in HR. Ends proficient for HR need to be covered by the recombinase Rad51. In yeast, one of Rad52s functions in HR is to mediate the exchange of RPA for Rad51. To investigate this step in HR, researchers analysed a truncated form of Rad52 (*Rad52-327Δ*). Normal Rad52 interacts physically with Rad51, but this mutant *Rad52* does not as the Rad51 interaction domain has been deleted. In the analysis, wild-type and *rad52-327Δ* mutant strains were transformed with either an empty plasmid (pRS426) or a plasmid overexpressing *RAD51* (pRAD51). In the latter strains, Rad51 contents are 100-fold higher than normal levels. The strains were tested for the ability to repair DNA DSBs produced by gamma irradiation, see Figure 1 below.

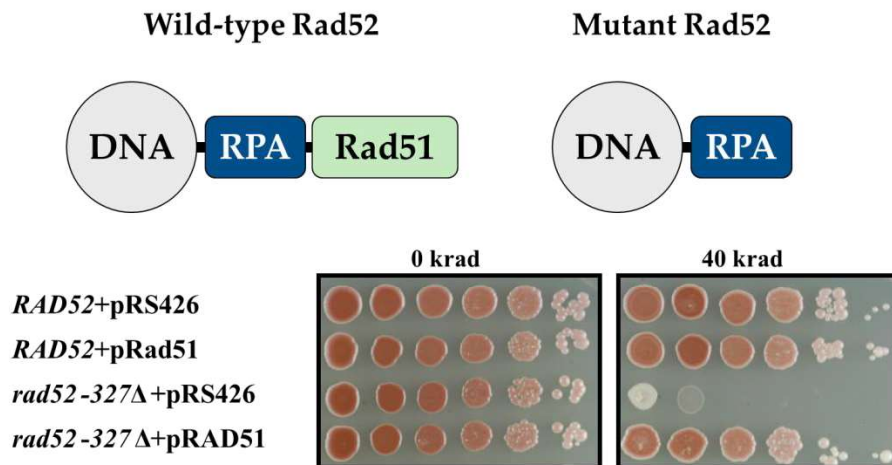


Figure 1. Wild-type and *rad52-327Δ* strains were transformed with either an empty control vector, pr426, or a vector over-expressing *RAD51* (pRAD51). 10-fold serial dilutions of cells for the four strains were spotted on solid medium selecting for the plasmids and exposed to either no gamma irradiation or 40 krad as indicated. For each strain the same dilution series of cells were used for both plates. The protein domain structures of Rad52 and mutant Rad52 are indicated in the top of the figure.

Question 1. Does gamma irradiation kill yeast cells if HR is defective? Explain your answer.

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Gamma irradiation *does* kill yeast cells when HR is defective, as seen by the strong sensitivity of the rad52-327Δ mutant. However, the experiment also shows that overexpressing Rad51 partially rescues the HR defect, allowing the mutant to survive radiation better than when Rad52's Rad51-interaction domain is missing and Rad51 is not overproduced. This indicates that very high Rad51 levels can bypass some—but not all—of the need for Rad52 during HR.

Esperanza's Answer:

Yes, it does. Compare strains RAD52+pRS426, which represents a wild-type situation, to rad52-327Δ+pRS426. The only difference between the two strains is the mutation in RAD52. With the wild-type around 50% of the cells survive irradiation. With the mutant strain no cells survive even when approximately 106 cells were spotted

Question 2. A strain that does not produce Rad52 at all shows a phenotype similar to the strain producing Rad52-327Δ. Such strains cannot be rescued by overproduction of Rad51. How can you use this information to evaluate whether the truncated protein Rad52-327Δ can fold *in vivo*.

You can use that comparison as an *in vivo* folding test for Rad52-327Δ.

- **A rad52Δ null strain makes no Rad52 protein at all. It is very sensitive to γ-irradiation, and this phenotype cannot be rescued by Rad51 overproduction.**
- **The rad52-327Δ strain has a *similar* sensitivity when Rad51 is at normal levels, but is rescued when Rad51 is overproduced.**

If Rad52-327Δ were badly misfolded or unstable *in vivo*, it would effectively be the same as the null: the cell would have no functional Rad52, and Rad51 overproduction would *not* help—just as in the rad52Δ strain.

The fact that Rad51 overproduction *does* rescue rad52-327Δ means that the truncated protein must still:

- **be stably produced and folded,**
- **retain other Rad52 functions (e.g. DNA/RPA binding, annealing, scaffold role),**

and only lack the specific Rad51-interaction domain that makes loading inefficient at normal Rad51 levels.

So, the difference between “no rescue” in rad52Δ and “rescue” in rad52-327Δ shows that Rad52-327Δ is a properly folded, partially functional protein *in vivo*, not just a misfolded or degraded fragment.

Esperanza's Answer:

Since strains producing Rad52-327Δ, unlike strains with no Rad52, can be rescued by Rad51; the truncated protein Rad52-327Δ must produce a folded protein that is partially functional

Question 3. Why does overexpression of *RAD51* rescue the phenotype of rad52-327Δ? Provide a simple (not in detail) model explaining this observation.

A simple model for why overexpression of RAD51 rescues the rad52-327Δ phenotype is based on suppression by mass action.

In the rad52-327Δ mutant, Rad52 cannot efficiently help Rad51 load onto RPA-coated single-stranded DNA, so at normal Rad51 levels homologous recombination rarely occurs and the cells are radiation-sensitive.

However, when RAD51 is massively overexpressed, the very high concentration of Rad51 allows it to bypass the defective mediator function simply by mass action: with so many Rad51 molecules

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present, enough of them can spontaneously replace RPA and assemble on ssDNA (Single-stranded DNA) even without proper assistance from Rad52.

Thus, suppression by mass action explains how excessive Rad51 partially restores HR and rescues the mutant phenotype.

Esperanza's Answer:

The Rad51 binding domain is used to attract Rad51 to Rad52 during repair. The Rad51 binding domain is not necessary at high concentrations of Rad51, simply because, at the high concentrations of Rad51, there will be Rad51 protein close to Rad52 at all times by chance. This is sometimes called suppression by mass action

ssDNA:

Single-stranded DNA (ssDNA) is DNA in which the two complementary strands have separated, leaving only one strand exposed. ssDNA is produced naturally during processes such as DNA replication, DNA repair, and homologous recombination.

Its key function in recombination and repair is that ssDNA serves as the platform on which repair proteins assemble. After a double-strand break, the DNA ends are resected to generate 3' ssDNA tails. These tails are first coated by the ssDNA-binding protein RPA and then replaced by Rad51, which forms a filament on the ssDNA that performs the homology search and strand invasion. Thus, ssDNA is the essential intermediate that allows the cell to find an intact template and repair broken DNA accurately.

Assignment 9. DNA replication

Task set A

Task set A contains 10 multiple choice questions (20 in final exam). The correct answer should be indicated by a letter in the cell next to the possible answers.

	Correct answer
1) Initiation of DNA replication of a eukaryotic chromosome requires:	
A) an RNA Primer B) a single-stranded DNA primer C) a double stranded DNA primer D) an rRNA A) an RNA primer — Correct. DNA polymerases cannot start DNA synthesis de novo; they require a free 3'-OH group, which is provided by a short stretch of RNA synthesized by primase (a short RNA primer).	
Why the other answers are wrong B) a single-stranded DNA primer — Incorrect. Although a single-stranded DNA molecule <i>could</i> in theory provide a 3'-OH, eukaryotic cells do not synthesize DNA primers. Primase is an RNA polymerase, and the replication machinery is built to use RNA, not DNA, for priming. C) a double-stranded DNA primer — Incorrect. A double-stranded primer offers no free single-stranded 3'-OH end for DNA polymerase to extend. Both strands are paired, making extension impossible without first unwinding or degrading part of it. Replication initiation requires a <i>single</i> free 3' end, not a base-paired duplex. D) an rRNA — Incorrect. rRNA is a structural and catalytic component of ribosomes, involved in translation, not replication. It is never used as a primer and cannot be recognized by DNA polymerase or primase for this purpose.	
2) Okazaki fragments are formed during synthesis of:	
A) the leading strand B) the lagging strand C) telomeres D) mRNA The correct answer is B) the lagging strand . Okazaki fragments arise because DNA polymerase can synthesize new DNA only in the 5'→3' direction. At a replication fork, one template strand is	

oriented so that DNA polymerase can move continuously toward the fork; this produces the leading strand, which therefore does not require Okazaki fragments. The other template strand runs in the opposite orientation, forcing DNA polymerase to synthesize DNA away from the replication fork in short bursts as the fork opens. These short, discontinuously synthesized pieces are the Okazaki fragments, later joined by DNA ligase to form a continuous lagging strand. The other options are incorrect for specific mechanistic reasons: • A) the leading strand—wrong because its template orientation allows continuous synthesis; no fragmentation is needed. • C) telomeres—wrong because telomere extension is performed by telomerase, an RNA-dependent DNA polymerase that adds repetitive sequences directly without forming Okazaki fragments. • D) mRNA—wrong because Okazaki fragments are unique to DNA replication, not RNA transcription; mRNA is synthesized by RNA polymerase as a continuous RNA transcript. Thus, Okazaki fragments are a direct consequence of the unidirectional activity of DNA polymerases and the antiparallel nature of DNA strands, and they define the discontinuous mode of lagging-strand synthesis.	
3) Okazaki fragments need to be synthesized because:	
A) RNA polymerases only have 5-OH to 3-OH polarity B) RNA polymerases only have 3-OH to 5-OH polarity C) DNA polymerases only have 5-OH to 3-OH polarity D) DNA polymerases only have 3-OH to 5-OH polarity The correct answer is C) DNA polymerases only have 5'-OH to 3'-OH polarity. Okazaki fragments are required because DNA polymerases can synthesize DNA only by adding nucleotides to the 3'-OH end of a growing strand, meaning synthesis proceeds exclusively in the 5' → 3' direction. At the replication fork, one template strand is oriented so that continuous 5' → 3' synthesis is possible (the leading strand). The opposite template strand, however, runs in the reverse orientation, forcing DNA polymerase to work away from the advancing fork. Since new template DNA becomes available only gradually, the polymerase must synthesize the lagging strand discontinuously, in short segments known as Okazaki fragments. These fragments are later processed and joined to form a continuous strand. The other answers are incorrect because they describe the properties of the wrong enzyme or an incorrect polarity entirely: • A) RNA polymerases only have 5'-OH → 3'-OH polarity—although RNA polymerases also synthesize 5' → 3', their polarity is not what determines the need for Okazaki fragments; replication discontinuity is dictated by DNA polymerase, not RNA polymerase. • B) RNA polymerases only have 3'-OH → 5'-OH polarity—false; RNA polymerases do not synthesize in the 3' → 5' direction.	